Colegio de Montalban

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Institute of Computing Studies

A Multi-Platform Learning Management System for Institute of Computing Studies at Colegio de Montalban

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CHAPTER 1 INTRODUCTION

1.1 BACKGROUND OF THE STUDY

LMS research connects course access with planning, monitoring, and reflection. Du et al. (2025) examined self-regulated learning support and reported stronger self-regulation, with an examination advantage on one test but not another. Çakiroğlu et al. (2024) related LMS interaction profiles to self-regulation. These findings support examining connected learning tools without assuming that access alone improves achievement.

Computing education also requires engagement mechanisms with clear instructional purposes. Ahmed et al. (2025) evaluated a gamified mobile/cloud LMS in an Object-Oriented Programming course and reported favorable achievement and engagement findings in that setting. Jaramillo-Mediavilla et al. (2024) reviewed gamification evidence and identified the need to examine motivation and performance together. Those findings justify investigating participation support but cannot be presumed for CdM LMS.

In the Philippines, studies show that the organization of an LMS affects how students and instructors experience academic tasks. Llamas et al. (2025) examined the Moodle-based LMS of Mindanao State University–Iligan Institute of Technology in Iligan City through discussions and task interaction involving six academics and six fourth-year students. They identified improvement areas in usability, communication, course building, monitoring and assessment, content management, and technical support. Laspiñas and Almerez (2024) surveyed 33 business graduate students at an unnamed university in Cebu City and reported generally neutral task-related and experience-related evaluations. The MSU–IIT study identified workflow concerns, while the Cebu study described users' overall experience. Both support examining clarity and usability through actual academic tasks instead of treating the availability of a platform as sufficient evidence of its suitability.

Within Colegio de Montalban, the present study focuses on the Institute of Computing Studies (ICS), including the academic workflows of its students and instructors and the administrative functions supporting them. Computing education requires coordination of learning resources, coursework, programming activities, assessment records, and faculty responsibilities. The research

group's initial project description identifies the use of separate tools for academic work and personal organization as a concern to be addressed. This description serves as the starting problem context, while the extent of fragmentation and users' specific difficulties remain matters for local requirements gathering.

The central development problem is how to organize these related activities within a shared environment while maintaining clear responsibilities and appropriate access to information. Students need understandable access to their coursework and academic records; instructors need manageable teaching workflows; academic supervisors need suitable course-level information; and administrators need controlled system-management functions. A connected design must also account for schedules, changes in class occurrences, and the privacy of personal academic information. Combining these responsibilities requires more than placing several tools on the same screen.

In response, this study proposes the CdM Learning Management System (CdM LMS), a multi-platform environment tailored to ICS. Its interconnected applications will provide complementary access to shared information, with the Web Portal serving as the principal academic environment, the Mobile Application supporting selected academic and productivity activities, and the Desktop Application serving as the Administrator Console. The study will develop and evaluate the system within the ICS setting. Its contribution will be evidence about the suitability and quality of this institutional implementation, rather than an assumed improvement in students' grades or long-term learning outcomes.

1.2 STATEMENT OF THE PROBLEM

This study seeks to determine the level of software quality of the LMS for ICS at Colegio de Montalban in supporting coordinated academic resource and grade management, learning support, academic coordination, and system administration. The evaluation will follow six selected characteristics of the ISO/IEC 25010:2023 Software Product Quality Model (International Organization for Standardization [ISO], 2023).

Specifically, the study seeks to answer the following questions:

1. What is the evaluated software quality of CdM LMS in supporting academic resource and grade-management workflows in terms of functional suitability, as measured by:

• Functional completeness in supporting course materials, assignments, dynamic grading, CodeLab activities, Git-integrated submissions, and the defined academic coordination functions;

• Functional correctness of weighted grade computations, EXP and level calculations, descriptive analytics, and schedule revisions and exceptions;

• Functional appropriateness in supporting connected academic workflows for BSIT and BSCpE users within their assigned roles and platform responsibilities?

2. What is the evaluated software quality of CdM LMS in providing understandable and engaging task interaction in terms of interaction capability, as reflected in:

• Appropriateness recognizability of course, grading, productivity, gamification, and academic coordination functions;

• Learnability of permitted tasks in the Web Portal, selected Mobile Application functions, and administrator-only Desktop Application;

• User engagement supported by EXP, levels, achievements, leaderboards, study tools, and the moderated Knowledge Exchange environment?

3. What is the evaluated software quality of CdM LMS in protecting academic information and controlling access in terms of security, as demonstrated by:

• Confidentiality of student grades, personal notes, academic records, and restricted course information;

• Integrity of grades, submissions, schedule records, and events during authorized updates and concurrent actions;

• Accountability through activity and audit records for designated administrative actions, academic coordination changes, and Knowledge Exchange moderation?

4. What is the evaluated software quality of CdM LMS in processing academic information and maintaining applicable shared records in terms of performance efficiency, as indicated by:

• Time behavior in computing grades, generating descriptive analytics and calendar views, retrieving course resources, and displaying relevant in-LMS updates;

• Resource utilization during the processing and retrieval of shared information by the authorized Web, Mobile, and Desktop components under defined workloads?

5. What is the evaluated software quality of CdM LMS in providing stable access to permitted academic and study functions in terms of reliability, as evaluated through:

• Availability of the Web Portal, supported mobile study tools, and Administrator Console during the defined evaluation period;

• Fault tolerance when processing interrupted requests, concurrent grade updates, or simultaneous Knowledge Exchange activity, including the handling of failed operations without incomplete or duplicate records?

6. What is the evaluated software quality of CdM LMS in supporting consistent information exchange and operation alongside other software in terms of compatibility, as indicated by:

• Interoperability among the Web Portal, Mobile Application, and Administrator Console through their applicable shared records and authorized backend workflows;

• Co-existence of the Administrator Console with other software in the supported administrator workstation environment without adversely affecting their operation?

1.3 OBJECTIVES OF THE STUDY

1.3.1 General Objective

To develop and evaluate a Learning Management System for the Institute of Computing Studies at Colegio de Montalban that coordinates academic management, learning support, academic oversight, and system administration through shared information and role-appropriate access.

1.3.2 Specific Objectives

1. Establish shared backend services and a centralized PostgreSQL database for the principal Web Portal, complementary Mobile Application, and administrator-only Desktop Application, with credential-based authentication, account-status controls, and server-enforced access for Student, Instructor, Super User, and Administrator roles.

2. Develop course and section management that allows authorized instructors to organize materials, annotations, assignments, announcements, and deadlines, and allows enrolled students to access resources and submit permitted work under the applicable validation rules.

3. Implement a Dynamic Gradebook with configurable assessment components and weights, score recording, automatic weighted-grade computations, personal student grade views, instructor performance summaries, and CSV grade export; verify computed values and charts against documented test records.

4. Provide CodeLab for in-browser programming execution and assessment against predefined test cases, and a separate Git-integrated workflow for repository-based coursework submission and instructor review; verify the supported submission and access rules for each workflow.

5. Develop notes, task lists, flashcards, review and mastery records, focus timers, and study-session tracking, together with rule-based EXP, levels, achievements, badges, streaks, and department leaderboards; verify progression calculations and the separation of EXP from course grades and forum reputation.

6. Implement Knowledge Exchange for academic questions, answers, and resource sharing, with authorized instructor moderation, Faculty-Verified Answers, contribution-based reputation, and records of designated moderation actions.

7. Develop the faculty directory and academic operations hub, distinguishing program affiliations and Dean or Program Head designations from primary account roles, and providing restricted course supervision that excludes student identities, individual grades, submissions, assessment attempts, and answers.

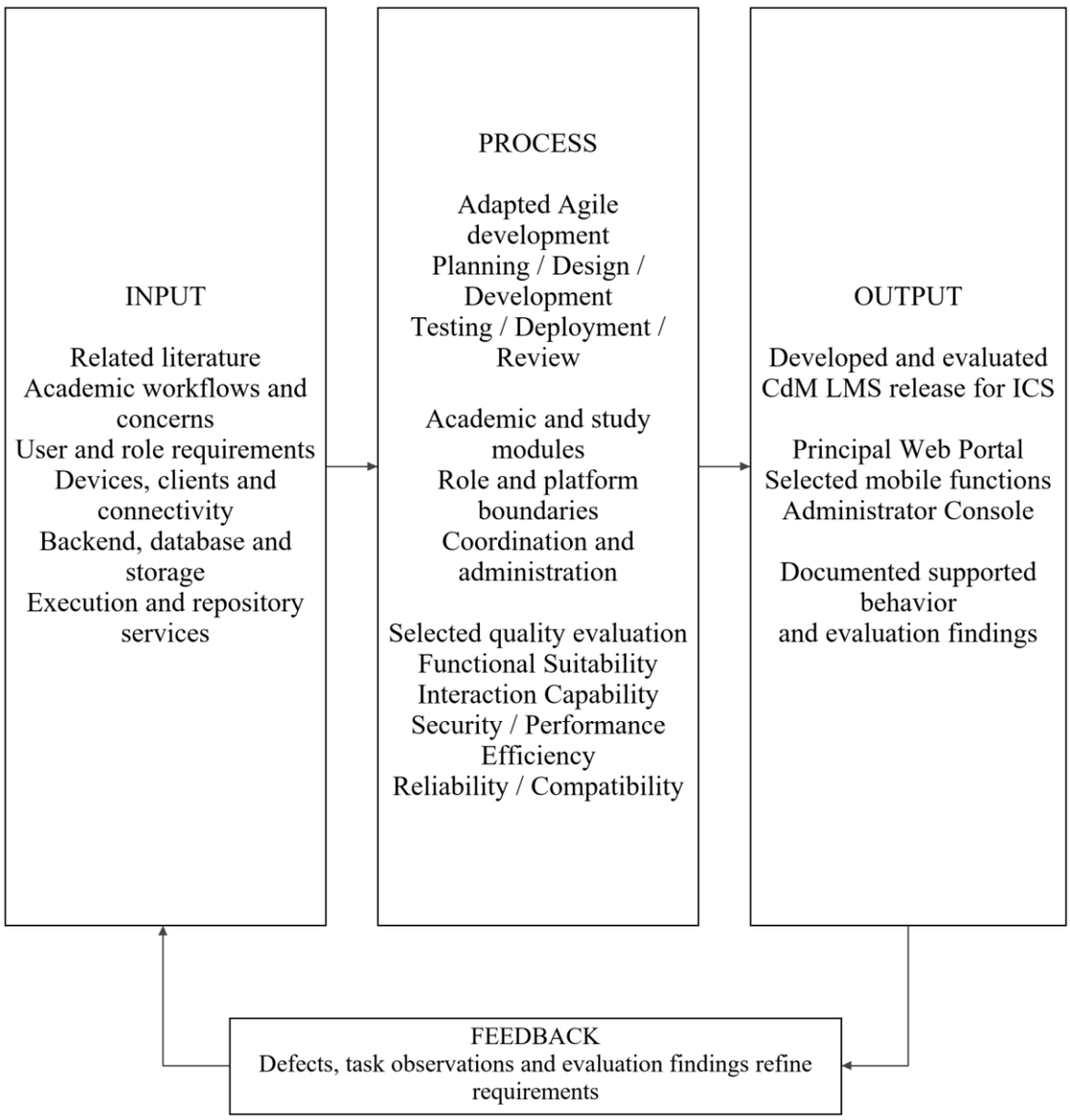
8. Implement Asia/Manila-based weekly course schedules with effective-dated revision history and future occurrence cancellation, rescheduling, and restoration; provide authorized calendar views, audience-targeted saved events, coursework deadlines, and private note–event snapshot conversion, and verify their visibility and preservation rules.

9. Provide the Administrator Console with account approval, status and role management, permitted configuration, activity and audit records, reporting, and the backup and recovery functions supported by the evaluated release; verify that administrative permissions remain separate from academic oversight.

10. Evaluate the developed system in terms of functional suitability, interaction capability, security, performance efficiency, reliability, and compatibility using role-appropriate tasks, feedback from eligible ICS students and instructors, and technical checks by qualified IT experts, and document defects and limitations identified in the evaluated version.

1.4 CONCEPTUAL FRAMEWORK

Figure 1. Conceptual framework of CdM LMS using IPO with feedback



The conceptual framework follows the Input–Process–Output (IPO) model with feedback, as shown in Figure 1. It organizes the study's requirements and resources, development and evaluation activities, and expected system output. Purwanto et al. (2025) examined limitations of linear IPO representations and proposed an IMOI adaptation in which outputs and feedback inform subsequent iterations. The present study retains IPO while incorporating feedback into requirement refinement and succeeding development cycles.

The Input component comprises related literature, initial academic practices and concerns within ICS, and system requirements. The literature covers academic management, programming

coursework, learning support, and collaboration. Local requirements gathering will examine the use of separate academic and productivity tools, resource management, faculty coordination, and access conditions. These concerns will be investigated rather than reported as established local findings in advance.

Technical inputs include development workstations, supported client devices, Next.js services, PostgreSQL, web browsers, the Expo mobile application, and the Electron Administrator Console. Connectivity, persistent storage, programming execution, repository access, authorization, continuity arrangements, and test configurations will define the operating environment. Estimated equipment specifications will be distinguished from configurations demonstrated during evaluation.

The Process component combines iterative development with requirements-based testing and software-quality evaluation. Planning, Design, Development, Testing, Deployment, and Review will organize successive increments. The modules will address academic management, computing coursework, personal study, gamification, collaboration, academic coordination, and administration. Their availability will follow Student, Instructor, Super User, and Administrator permissions and the complementary responsibilities of the three applications.

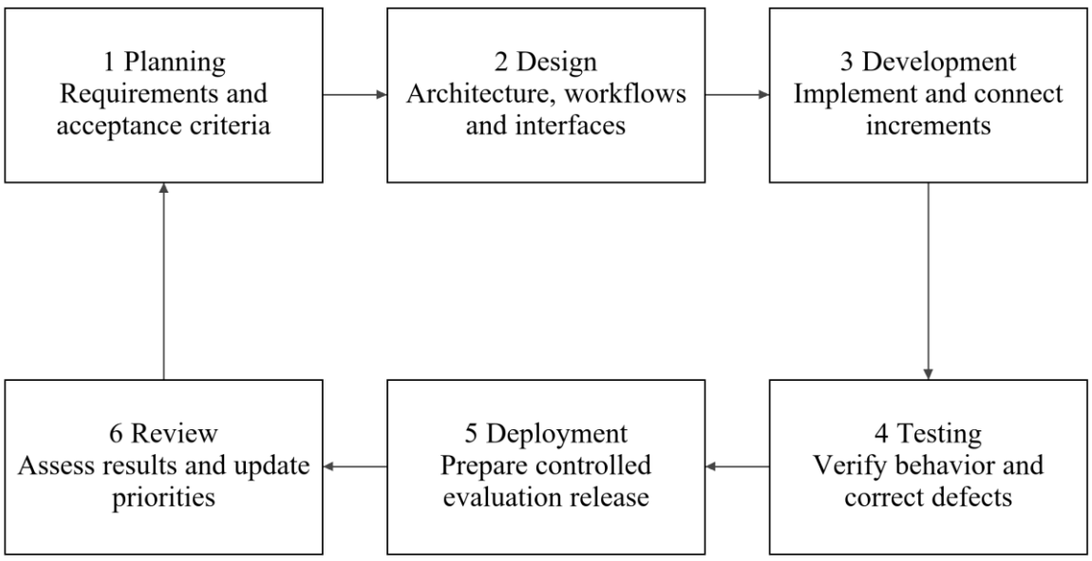
The evaluation will use six selected ISO/IEC 25010:2023 characteristics: Functional Suitability, Interaction Capability, Security, Performance Efficiency, Reliability, and Compatibility (ISO, 2023). Respondent ratings will describe experiences with assigned tasks, while technical checks will examine calculations, access restrictions, record handling, performance, and shared information. Abid et al. (2024) provide a feature-based example of systematic LMS evaluation; the study's quality characteristics are derived from ISO/IEC 25010:2023.

The Output is the developed and evaluated CdM LMS release for ICS. It will provide a principal Web Portal, selected mobile support, and a native Administrator Console around shared records and authorized workflows. Feedback from testing, evaluation, and review will return to the Input component to refine requirements and priorities. Revised inputs will guide subsequent

development cycles. Findings will concern the evaluated release and will not establish long-term educational effects or institution-wide adoption.

1.5 OPERATIONAL FRAMEWORK

Figure 2. Adapted Agile development cycle for CdM LMS



The researchers will use an adapted Agile development approach consisting of Planning, Design, Development, Testing, Deployment, and Review. Alsaqqa et al. (2020) describe Agile values, iterative methods, and responsiveness to changing requirements. The six stages used here are the project's operational organization rather than a universal Agile sequence. Each cycle will address prioritized requirements and produce an increment for examination and refinement.

Planning Phase. The researchers will gather local requirements from ICS students, instructors, and relevant academic and administrative personnel. Llamas et al. (2025) identify LMS usability and workflow concerns, while Guico (2026) highlights preparation, resource access, and training needs. These findings inform topics for local investigation. Requirements will specify role responsibilities, platform coverage, expected behavior, and acceptance criteria in a prioritized backlog.

Design Phase. Architecture, database relationships, workflows, and interfaces will be prepared from the requirements. Flowcharts, ERDs, and DFDs will identify information movement and component responsibilities. Attention to navigation and task feedback is supported by the user-

experience concerns examined by Laspiñas and Almerez (2024). The design will separate academic supervision from teaching and administration and protect personal academic records.

Development Phase. Selected functions will be implemented incrementally using the Next.js backend, PostgreSQL database, and applicable clients. Development will connect academic management, programming coursework, study tools, gamification, collaboration, coordination, and administration. Each increment will be checked against its acceptance criteria. The release record will distinguish implemented, proposed, tested, and evaluation-enabled functions.

Testing Phase. Controlled accounts and datasets will be used to examine correctness, authorization, integrity, integration, performance, and failure handling. The evaluation release will also undergo role-appropriate task assessment aligned with the six selected ISO/IEC 25010:2023 characteristics (ISO, 2023). Technical results and respondent ratings will be reported separately. Defects will be corrected and retested before affected functions are evaluated.

Deployment Phase. A reviewed release will be configured in a controlled environment. The researchers will verify startup, backend connectivity, accounts, client builds, and available continuity procedures. Participants will receive task guidance. The release version, enabled modules, device configurations, and unresolved limitations will be recorded; controlled evaluation will not be described as institution-wide deployment.

Review Phase. The researchers will examine technical observations, questionnaire responses, and task feedback from eligible ICS users and qualified IT experts. Findings will identify supported behavior, defects, and refinement priorities. These priorities will return to the backlog for the next cycle, and corrected functions will undergo further testing.

1.6 SCOPE AND LIMITATIONS

The study covers the development and software-quality evaluation of CdM LMS for the Institute of Computing Studies at Colegio de Montalban, including BSIT and BSCpE students, instructors, and supporting academic and administrative workflows. Participation is not restricted to

one student year level. IBE and ITE configuration is a technical application capability; those institutes are outside the research population and generalization of findings.

The Web Portal is the principal academic environment and the initial platform for the coordination expansion. Mobile provides selected academic and productivity functions according to the evaluated build. The native Desktop Application is the Administrator Console. Super User oversight is web-based; native mobile parity for the expansion and native Super User console entry are excluded. A role and platform matrix will identify the functions actually tested.

Functional coverage includes course resources and submissions, the Dynamic Gradebook, descriptive analytics, separate CodeLab and Git coursework workflows, Knowledge Exchange, personal notes and study tools, gamified progression, faculty organization, restricted supervision, schedules, targeted events, and administration. Dean and Program Head are academic designations attached to eligible accounts. They do not replace the four primary roles or automatically grant Super User privileges.

Evaluation will use criterion-based purposive sampling of eligible ICS students and instructors with prior LMS experience and qualified IT experts. Participants must voluntarily agree and perform relevant tasks before rating them. Full eligibility, recruitment, task assignment, and sample justification procedures are specified in Chapter 3. Validator participation is separate from respondent participation.

Operation depends on supported devices, connectivity, and service availability. The study does not claim universal operating-system support, identical platform functions, uninterrupted service, or full offline coursework. Hardware figures are estimates until tested. Programming languages, repository access, reporting, and continuity coverage will be limited to the documented evaluation release.

Academic analytics describe existing records. Rule-based attention indicators do not predict future achievement. CodeLab checks predefined cases, while Git history supports review without proving authorship or eliminating plagiarism. Tab switching, copy/paste, and fullscreen events are

contextual assessment indicators rather than conclusive cheating evidence. Optional grade incentives will be evaluated only when their effect and institutional authorization are documented; EXP, forum reputation, and academic grades remain distinct measures.

Scheduling uses Asia/Manila, selected weekdays, one shared time pair, date bounds, and effective-dated revisions. Individual future classes may be cancelled, rescheduled, or restored. Arbitrary recurrence, multiple weekly time patterns, automatic attendance, holiday cancellation, room-conflict prevention, and timetable optimization are excluded. Student recurring timetable projection is excluded; authorized events and coursework deadlines remain available.

Supervision excludes student identities, individual grades, submissions, attempts, answers, and teaching controls. Event audiences govern LMS visibility without implying external delivery or Google synchronization. Note–event conversion creates private snapshots; later edits do not propagate. The descriptive study will not determine long-term changes in achievement, motivation, employability, or study habits. Findings apply to the selected ICS participants, tasks, and evaluated release.

1.7 SIGNIFICANCE OF THE STUDY

The proposed CdM LMS may contribute to the Institute of Computing Studies (ICS) by providing an integrated environment for managing academic and technical learning activities across Web, Mobile, and Desktop platforms. The study may provide value to the following:

Institute of Computing Studies and Colegio de Montalban. The study may provide an institutional basis for organizing related academic information and examining whether a tailored LMS meets the needs of ICS. Its documented requirements and evaluation findings may support decisions about further development, maintenance, and possible adoption.

Students. The system is intended to provide a coordinated place for accessing coursework and feedback and managing personal study activities. Students may benefit from clearer access to their own academic information and opportunities for moderated peer support. These expected benefits remain distinct from measured improvements in academic achievement.

Instructors. The study may support more consistent handling of course resources, submissions, grading records, programming activities, and academic communication. Evaluation findings may identify interaction or workflow difficulties that require correction before broader use.

Super Users and Program Heads. Restricted Super User supervision may support authorized academic oversight. Program Heads may benefit from faculty information and designation-specific event publication. These responsibilities will remain distinct from teaching and system administration.

Administrators. The dedicated console is intended to support controlled account operations, system records, reporting, and supported continuity functions. Documenting access boundaries and audit behavior may assist in maintaining accountability for administrative actions.

Future researchers and developers. The study may provide a reference for examining an institutional LMS with complementary applications, computing-specific workflows, and limited academic supervision. Its limitations and evaluation findings may also identify questions requiring separate research, including longer-term educational outcomes and broader platform coverage.

1.8 GENERAL FEATURES OF THE SYSTEM

CdM LMS comprises complementary applications connected through shared backend services and a centralized database. The following groups describe intended feature coverage. A release record will distinguish existing modules, planned extensions, tested functions, and functions enabled for evaluation. Availability is governed by role and platform coverage.

Academic and Course Management

Authorized instructors organize courses, sections, learning materials, annotations, assignments, announcements, and deadlines. Students request enrollment using valid join codes and obtain course-workspace access after approval. Streams, classwork, people, and grades present information according to enrollment and course permissions.

Grading and Descriptive Analytics

The Dynamic Gradebook uses configured assessment components, weights, and recorded scores to compute grades. Students view their own grade information; instructors review authorized class summaries and export permitted CSV records. Rule-based attention indicators use available activity and performance records. Computation and chart outputs will be verified against controlled data.

Programming Coursework

CodeLab provides browser-based code writing, execution, and submission assessed against instructor-defined test cases. Git-integrated coursework separately connects authorized repository submissions to instructor review. Repository authorization remains separate from LMS authentication. Execution languages, service configuration, and supported submission paths will be recorded for the evaluation release.

Personal Study and Productivity

Owned notes, personal tasks, authorized calendar items, flashcards, review/mastery records, focus timers, and study-session tracking support organization and repeated study. Users may schedule an owned note as a private event or save a readable persisted event as a private note. Conversion reuses linked results and creates snapshots rather than continuous synchronization.

Gamification and Collaborative Learning

Configured participation earns EXP, levels, achievements, badges, streaks, and permitted rankings. Knowledge Exchange supports academic questions, answers, and resource sharing, with instructor moderation and Faculty-Verified Answers. Forum reputation is separate from EXP and grades. Reward or incentive rules will be documented and evaluated only within their approved scope.

Academic Organization and Oversight

The ICS directory and organizational chart identify the Dean, BSIT and BSCpE Program Heads, and faculty memberships. General means neither track; Cross-program means both. Course program classification remains independent of faculty membership. Super Users use the web

academic hub to inspect authorized summaries, faculty information, timetables, and restricted course supervision, and make narrowly permitted leadership assignments.

Schedules and Calendar Coordination

Authorized instructors and Administrators manage date-bounded weekly schedules and occurrence exceptions. Effective revisions preserve earlier times, rooms, and instructor attribution. Day, Week, Month, and Year views distinguish saved events, projected classes, and syllabus deadlines; Year uses density summaries. Explicit audiences include Personal, Course, Institute Faculty, BSIT Instructors, and BSCpE Instructors. Program Heads may publish to the documented ICS audiences without gaining ownership of other creators' content.

Administration and Information Protection

The native Administrator Console provides account status and role operations, instructor assignments, configuration, logs, and supported reporting and continuity functions. Only Administrators provision or revoke Super User accounts. The backend checks sessions and applicable role, institute, ownership, enrollment, and resource relationships. Protected operations use validation, audit records, and the transactions or version checks specified by the feature contract.

Platform Integration and Assistance

The clients retrieve applicable shared information through authorized Next.js workflows. Updates become available according to implemented retrieval or synchronization behavior and connectivity. In-LMS notifications and the platform help chatbot support navigation and academic tasks. Feature presence does not establish universal native parity, external notification delivery, or an AI tutoring function.

Table 1. Role and platform responsibilities

Role	Principal interface	Permitted responsibility	Boundary
Student	Web; selected mobile functions	Approved coursework, own grades, study tools and collaboration	No faculty directory or supervision
Instructor	Web; selected mobile functions	Assigned teaching, grading, enrollment and schedules	Program Head capabilities require designation

Role	Principal interface	Permitted responsibility	Boundary
Super User	Web academic operations hub	Safe supervision and narrow faculty coordination	No grading, assessments, schedule editing or system administration
Administrator	Native Desktop Console	Protected account/configuration, audit and supported continuity operations	Additional browser/mobile administration is outside this evaluation scope

1.9 DEFINITION OF TERMS

Academic Calendar - The interface presenting authorized saved events, generated class occurrences, and coursework deadlines through Day, Week, Month, and Year views.

Academic Operations Hub - The web interface through which a Super User views authorized academic summaries, faculty information, timetable filters, and course-supervision links.

Administrator - The system role responsible for authorized account, permission, configuration, log, reporting, and continuity operations through the native Administrator Console.

Agile Development - The iterative development approach in which requirements are implemented, tested, reviewed, and refined through successive project cycles.

Audit Log - A record of designated actions and changes used to support accountability, including authorized administrative and academic-leadership operations.

BSCpE - Bachelor of Science in Computer Engineering, one of the ICS programs within the study context.

BSIT - Bachelor of Science in Information Technology, one of the ICS programs within the study context.

CdM LMS - The proposed interconnected learning management environment developed for the Institute of Computing Studies at Colegio de Montalban.

Centralized Database - The shared PostgreSQL data store supporting authorized academic and system records used by the applicable applications.

CodeLab - The in-browser programming environment for writing, executing, and submitting code assessed against predefined test cases.

Course Supervision - Restricted, read-only access to course metadata, instructor attribution, timetable, announcements, syllabus, and learning-material summaries.

Dean - An academic leadership designation represented in the institute directory; the designation itself does not grant additional system permissions.

Dynamic Gradebook - The grading module that uses configured components, weights, and recorded scores to compute and display authorized grade information.

Effective Date - The date from which a particular timetable revision governs the generation of applicable recurring class occurrences.

Experience Points or EXP - Configured participation points used for gamified progression and related rewards; they are not equivalent to course grades or forum reputation.

Faculty Directory - The authorized listing of ICS faculty, program affiliations, leadership designations, and relevant assigned-course summaries.

Faculty-Verified Answer - A Knowledge Exchange response marked by authorized faculty through the platform’s answer-verification process.

Flashcard Review - Repeated study of saved question-and-answer cards using the review and mastery functions available in the evaluated release.

Gamification - The use of configured points, levels, badges, achievements, streaks, and leaderboards to support participation in permitted LMS activities.

Git-Integrated Submission Engine - The workflow for repository-based coursework submission and authorized instructor review of applicable programming activities.

Institute of Computing Studies or ICS - The institute at Colegio de Montalban that defines the development context and research-evaluation scope of the study.

Instructor - The system role responsible for authorized teaching, assessment, grading, and course-related communication for assigned courses.

Knowledge Exchange - The moderated environment for academic and technical questions, answers, and resource sharing, with faculty verification and reputation mechanisms.

Learning Management System or LMS - The application environment used in this study to organize course resources, coursework, assessment, academic records, and related support workflows.

Program Head - An academic designation attached to an eligible ICS Instructor account for a specified program, with defined coordination privileges.

Reputation Points - Configured points associated with Knowledge Exchange contributions, maintained distinctly from EXP unless an explicit rule connects them.

Role-Based Access Control or RBAC - The mechanism that restricts system capabilities according to assigned roles, supplemented by institute, ownership, enrollment, and related-resource checks.

Schedule Exception - A record modifying an individual class occurrence through cancellation or rescheduling, or supporting restoration of its original occurrence.

Snapshot Conversion - Creation of a private note or calendar event from accessible source content at the time of conversion, without continuous synchronization.

Student - The system role permitted to access enrolled-course content, submit assigned work, view authorized personal academic records, and use available learning-support functions.

Super User - The system role for limited academic oversight and coordination, separate from Instructor and Administrator privileges.

Targeted Event - A saved academic event whose visibility and management are governed by its designated personal, course, institute-faculty, or program-instructor audience.

Timetable Revision - An effective-dated version of a course schedule used to preserve the historical representation of recurring class times.

Visual Grade Analytics - Descriptive charts and summaries derived from available grade and performance records, without predictive or AI-based inference.

CHAPTER 2 REVIEW OF RELATED LITERATURE

Digital technology has changed how educational institutions organize instruction and manage academic information. Learning Management Systems (LMS) provide a shared environment for distributing course materials, managing assignments, recording grades, and communicating with students. Access through different devices also creates opportunities to connect classroom activities with independent study. These functions provide the foundation for examining how an LMS can support academic work within higher education.

In the Philippine context, the studies reviewed in this chapter examine LMS adoption, user experience, accessibility, and institutional support. They also explore programming practice, gamification, personal learning strategies, and online collaboration. Their findings provide different perspectives on digital learning, including the difficulties users encounter and the conditions that support meaningful participation. Understanding these findings helps guide the selection and integration of functions appropriate to the Institute of Computing Studies (ICS) at Colegio de Montalban.

The project developers reviewed scholarly articles, conference papers, and system-based studies to establish the foundation for the proposed Learning Management System. Philippine and international studies are organized under eight topic groups: Multi-Platform Learning Management Systems; Course Management and Academic Communication; Programming Practice and Automated Assessment; Git-Based Coursework Submission; Gamification and Student Engagement; Personal Study Tools and Self-Regulated Learning; Online Collaboration and Knowledge Sharing; and Academic Coordination and Information Boundaries. The discussion compares their findings, explains their connection to the proposed system, and identifies the research gap that the study will address.

2.1 Related Literature

2.1.1 Multi-Platform Learning Management Systems

Access through different devices should support useful academic tasks and an understandable learning environment. Alfalah (2023) surveyed 258 students from Hail University, Aljouf University,

and Northern Border University in Saudi Arabia about mobile LMS adoption, including the Blackboard application. Performance expectancy, effort expectancy, and learning intention were associated with behavioral intention to use the system. In the Philippines, Dela Rosa (2023) examined Canvas Free for Teachers with 132 third-year BSIT students taking Web Systems and Technologies 2 at Bulacan State University. Students evaluated its lesson modules and assessments favorably. Both studies connect adoption with users' experiences of academic functions, although Alfalah examined mobile-use intentions and Dela Rosa examined acceptance within a specific programming course.

Philippine development studies provide more direct examples of designing learning access across devices. Quimod et al. (2023) developed e-Modyul, a mobile and web LMS for Technical-Vocational-Livelihood education, and evaluated it with 10 teachers and 20 students. Its approach supports accessing lessons and assessments through different operating systems. Ramos et al. (2024), researchers associated with the University of the Philippines Open University (UPOU), developed a Filipinized LMS responsive to Filipino nursing educators and learners. Their design considered connectivity limitations, multimodal learning materials, communication, and collaboration. These studies share an emphasis on access, but e-Modyul focuses on device coverage, while the Filipinized LMS emphasizes adaptation to its users and educational context.

Platform access also depends on the quality of everyday workflows. Llamas et al. (2025) examined the Moodle-based LMS at Mindanao State University–Iligan Institute of Technology (MSU–IIT), involving six academics and six fourth-year students. They identified concerns in usability, communication, course building, monitoring and assessment, content management, and technical support. Their findings qualify the favorable acceptance evidence from Dela Rosa: providing an LMS does not remove all difficulties in using it.

Taken together, the reviewed studies support designing complementary interfaces around the tasks and access conditions of the intended users. For CdM LMS, the web application will provide the principal academic environment, the mobile application will support selected student and instructor functions, and the desktop application will serve the Administrator. Multi-platform access therefore concerns coordinated responsibilities and consistent shared records rather than identical

functionality on every device. The reviewed studies inform this arrangement, while its suitability for ICS will require evaluation of the implemented applications.

2.1.2 Course Management and Academic Communication

Course management connects learning materials, activities, academic records, and communication within a clear workflow. Peralta (2024) developed a course management system for Centro Escolar University and evaluated it with 100 students and eight faculty members. It organized materials, announcements, assignments, quizzes, and grades and included Short Messaging Service notifications. Llamas et al. (2025), however, identified course-building, content-management, communication, and assessment concerns in the MSU–IIT LMS. These studies suggest that organizing functions within one platform is useful, but users must still be able to complete their tasks without unnecessary difficulty.

The communication experience also varies across educational settings. Acala and Dimasacat (2023) examined perceptions of 74 private senior high school students in Manila and reported favorable views of LMS-supported interaction and learning. They recommended training to help students and teachers use the available functions. Laspiñas and Almerez (2024), in contrast, surveyed 33 business graduate students at an unnamed university in Cebu City and reported generally neutral evaluations of the LMS's pragmatic and hedonic qualities. Their findings address different populations, yet both indicate that communication and task support should be assessed through users' experiences rather than inferred from the presence of features.

Institutional support affects whether these workflows can be maintained. Guico (2026) examined 182 faculty members and 29 administrators from 11 higher education institutions in Batangas, including Batangas State University, Lipa City College, and Rizal College of Taal. Reported concerns included preparation time, resistance to change, access to information and communication technology resources, and continuing training. These findings place course management within faculty support and institutional coordination.

Across the reviewed studies, effective academic communication depends on understandable course organization, appropriate user preparation, and the support needed to sustain the system. This evidence informs CdM's course materials, announcements, submissions, feedback, dynamic grading, and descriptive grade analytics. It also supports considering faculty organization, schedules, and targeted academic events as part of course coordination. Peralta's SMS function remains a comparison feature of that system; the proposed CdM workflow uses its own documented notification functions. Requirements gathering at ICS will establish the local difficulties to be addressed.

2.1.3 Programming Practice and Automated Assessment

Programming practice requires suitable exercises and feedback that explains performance against clear criteria. Messer et al. (2024) reviewed 121 papers on automated grading and feedback tools for programming education. Many tools evaluated correctness through dynamic methods such as unit tests, while readability, maintainability, and documentation received less attention. Novak and Kermek (2024), from the University of Zagreb in Croatia, examined automation for complex programming assignments using unit testing and custom scripts. Both studies support automating selected checks while retaining instructor judgment for aspects that those checks cannot adequately assess.

Philippine research adds considerations about introductory practice and access. Asuncion et al. (2022), affiliated with Ateneo de Manila University and the National Olympiad in Informatics–Philippines, presented the ABC Workbook and AutoJudge++. The project combines introductory Python exercises with an offline desktop checker that compares outputs against test cases. Classroom effectiveness evaluation remained future work. Arada (2025) examined Tinkercad with 42 first-year Computer Technology students at a Philippine state university. The platform received a below-average System Usability Scale score of 58, despite many students expressing willingness to use it. These studies concern different tools, but together they show that opportunities to practice should be considered alongside exercise design, usability, and access conditions.

Assessment can also concern development processes beyond the output of an individual program. Fernandez-Gauna et al. (2023), from the University of the Basque Country in Spain,

described automated feedback and assessment for team-coding assignments in a DevOps context. Their approach used team and individual indicators and periodic reports to support monitoring. Compared with the test-case approaches examined by Messer et al. and Asuncion et al., this work illustrates a broader form of assessment involving collaborative development activity. It does not establish that development indicators can replace assessment of the submitted program.

The reviewed studies support matching automated feedback to the assignment and preserving the limits of what is measured. CdM's CodeLab will evaluate programming work against predefined test cases and provide understandable results. Its checks will concern documented expected behavior rather than complete program quality. Git-based coursework review will remain a separate workflow. The related studies therefore guide test design, feedback presentation, and usability without expanding CodeLab into automated code repair, artificial intelligence grading, or DevOps-based team assessment.

2.1.4 Git-Based Coursework Submission

Repository-based coursework can connect code submissions with an organized record of development and instructor review. Schauer et al. (2024), from Heriot-Watt University in the United Kingdom, developed a framework integrating Canvas LMS and GitLab. They identified fragmented information and poor user experiences when the platforms were used independently and implemented eight applications to connect their workflows. Orvalho et al. (2024), affiliated with the University of Lisbon and Czech Technical University in Prague, presented GitSEED, a GitLab-backed automated assessment tool. Both approaches connect academic work with version control, although Schauer et al. emphasize LMS integration and GitSEED provides assessment and feedback through repository workflows.

Evidence about students' use of repositories complements these system-development approaches. Patani et al. (2024) examined GitHub use in a software engineering course with 319 undergraduate students, using surveys and repository records such as commits, pull requests, and issues. They reported improvements in collaborative development and teamwork after the course experience. Fernandez-Gauna et al. (2023) used repository-related development indicators to provide

feedback on team-coding assignments. Together, these studies show how repositories can make development activity available for educational review, but their measures extend beyond recording a final coursework submission.

The reviewed studies support examining how academic instructions, repository records, and instructor feedback can be connected. For CdM LMS, Git authorization will support the documented repository submission and review process, while LMS account access will follow its separate authentication and approval rules. Repository submissions will not automatically become CodeLab assessments or commit-based academic grades. The project's contribution is to evaluate its particular coursework workflow for ICS, informed by the integration concerns identified by Schauer et al. and the educational repository use described in the other studies.

2.1.5 Gamification and Student Engagement

Gamification can encourage participation when rewards are connected to meaningful learning activities. Ahmed et al. (2025) evaluated a gamified, mobile, cloud-based LMS with 60 second-year students taking Object-Oriented Programming at Mansoura University in Egypt. The experimental group obtained higher post-test achievement and engagement scores than the comparison group. Jaramillo-Mediavilla et al. (2024) synthesized gamification research and reported favorable findings in parts of the literature while identifying gaps in examining motivation and academic performance together. These studies support investigating gamification, but the outcomes of a particular intervention should not be treated as a guaranteed effect of every reward system.

The design of participation incentives matters. Jack (2025), in a conference abstract, discussed points and leaderboards in a flipped statistics classroom and considered benefits for motivation and engagement alongside the risks of superficial engagement and demotivation. Philippine evidence from Fojas (2022) examined gamification in Moodle courses at UPOU, comparing a gamified cohort with earlier non-gamified cohorts and collecting student feedback. The reported differences favored the gamified course, but forum-grading arrangements also differed between cohorts, and some students described anxiety or limited interest in rewards. Both studies show why participation counts and positive feedback need to be interpreted in relation to the design of the activities.

Benben and Bug-os (2022) studied Quizizz-based formative assessment with Grade 10 physics students at Central Mindanao University Laboratory High School in Bukidnon. Achievement and motivation increased after the intervention, although the reported learning gain was low. Its one-group pretest-posttest design and secondary-school setting differ from Ahmed et al.'s computing-course comparison. The studies therefore provide complementary evidence about possible motivational benefits while leaving questions about transfer to other subjects and populations.

Across the reviewed studies, gamification is most relevant when its rules encourage participation without obscuring academic expectations. CdM LMS will use EXP, levels, achievements, streaks, and leaderboards under defined rules. These records will remain separate from academic grades and Knowledge Exchange reputation. The literature supports the rationale for these mechanisms and the need to examine users' experiences, while the proposed descriptive software-quality evaluation will not establish their long-term causal effects on achievement or motivation.

2.1.6 Personal Study Tools and Self-Regulated Learning

Personal study tools can support planning, active review, and reflection when learners use them purposefully. Du et al. (2025) examined an LMS enhanced with self-regulated learning support among 69 management undergraduates taking mathematics at an unnamed large public university in China. Students reported stronger self-regulation, but an examination advantage appeared on the first test and not the second. Heikkinen et al. (2023), in a review of 56 studies, found that only 46 percent of the reviewed learning-analytics interventions reported positive effects on learning. Both studies indicate that learning support deserves investigation, while access to monitoring tools alone does not ensure improved outcomes.

Students' interactions with LMS tools help explain differences in how learning support is used. Çakiroğlu et al. (2024) examined a third-year Operating Systems and Applications course at an unnamed public university in Turkey. They identified different learner profiles and relationships between assessment interactions and self-regulation skills, including goal setting, self-evaluation, and help-seeking. In the Philippines, Funa et al. (2023) surveyed 301 preservice teachers in the Bicol region and supplemented the survey with interviews. Participants reported relatively strong

environment structuring and goal setting but also difficulties with task strategies, time allocation, priorities, and interaction. These studies connect self-regulation with both individual learning practices and the conditions in which academic tasks are completed.

Active review provides another form of personal learning support. Pan et al. (2022) compared user-generated and premade digital flashcards in six experiments involving delayed tests. Creating flashcard content generally supported memory and, in several conditions, application performance. Unlike the broader LMS and analytics studies, this research examines a specific review activity. It offers a direct basis for considering student-created flashcards alongside the organization and reflection functions discussed by Du et al., Çakiroğlu et al., and Funa et al.

Together, the reviewed studies support making planning, review, and progress information available within the learning environment. For CdM LMS, personal notes, tasks, flashcards, review and mastery records, and focus-study records will provide ways to organize and revisit academic work. Descriptive displays can help users inspect their saved records, but their presence will not be treated as evidence of improved self-regulation. The implemented functions will require evaluation for usability and correct behavior, while their educational effects remain a possible direction for later research.

2.1.7 Online Collaboration and Knowledge Sharing

Online knowledge sharing depends on the quality of interaction and the way discussions are organized. Oyarzun and Martin (2023) reviewed 63 studies on online learner collaboration and examined technologies, instructional design, facilitation, and outcomes. Their review connects collaboration with the conditions supporting participation. Bach and Thiel (2024) surveyed 298 undergraduate teacher-education students in asynchronous university courses and examined cognitive, metacognitive, and relational dimensions of interaction. Their findings concern associations with self-reported learning gains and satisfaction rather than causal effects. Both studies indicate that a shared online space should be assessed through the exchanges it supports.

Structured discussions and instructor involvement provide more specific guidance. Mudau and Van den Berg (2023), from the University of South Africa, examined online discussions involving 11 master's students in education. They identified the importance of clear guidelines, structured questions, and opportunities for discussion within a community of inquiry. Almodiel (2022) analyzed knowledge construction in asynchronous discussions within a graduate colloquium at UPOU. The study used the Interaction Analysis Model to examine the exchanges through which learners shared and developed knowledge. These studies support considering discussion structure and facilitation alongside opportunities for students to contribute.

Participation measures should also reflect interaction between users. Kumi (2023) examined discussion-forum exchanges in an undergraduate class with 31 students using social network analysis. The analysis revealed structural and relational aspects of participation that simple counts or self-reports could overlook. This complements Almodiel's examination of knowledge construction: both approaches look beyond the number of messages to consider how students interact. Their analytical methods are research approaches rather than functions automatically required in the proposed CdM system.

The reviewed studies support designing Knowledge Exchange as an organized space for academic questions, answers, and resource sharing. Moderation and Faculty-Verified Answers will provide project-specific ways to manage contributions and identify instructor-reviewed responses. Reputation will describe forum participation under its configured rules and remain separate from EXP and academic grades. The literature provides reasons to examine participation, facilitation, and contribution quality, while the suitability of CdM's verification and reputation mechanisms will require its own evaluation. The proposed forum supports asynchronous exchanges; the reviewed studies do not require adding live meetings or collaborative document editing.

2.1.8 Academic Coordination and Information Boundaries

Institutional LMS support extends beyond individual coursework. Guico (2026) identifies faculty preparation, resource access, and continuing training as management concerns. Llamas et al. (2025) identify monitoring, assessment, communication, and course organization as user-experience

concerns. Together, these findings provide reasons to examine faculty coordination and access to appropriate course information. They do not validate the particular schedule schema or supervision policy proposed for CdM LMS.

The project's academic coordination requirements will therefore be established through local consultation and explicit role policies. Directory information, timetable revisions, event audiences, and restricted supervision will be tested against those requirements. Academic designation and technical privilege will remain separate; shared calendar visibility will not imply access to private notes or individual assessment records.

2.2 Supporting Quality Framework and Local Study Context

ISO/IEC 25010:2023 provides the product-quality model supporting the study's selected evaluation characteristics: functional suitability, interaction capability, security, performance efficiency, reliability, and compatibility (International Organization for Standardization, 2023). The standard supports the quality criteria used in this study. Questionnaire feedback will describe respondents' evaluations, while calculation checks, access tests, and integration verification will provide evidence about implemented behavior.

The reviewed Philippine studies inform the project's computing-course, LMS, programming, and collaborative workflows. Their findings do not establish conditions within ICS. Local requirements gathering will identify the needs and experiences of the intended users.

2.3 Related Systems

The following systems provide documented examples of course organization, progress information, and repository-based workflows. The comparison uses the capabilities described in the cited sources rather than an exhaustive feature audit. It does not assume that these platforms lack other functions or cannot be extended.

2.3.1 Moodle

Moodle's publication on LMS 4.5 described improvements to assignment interaction and course subsections for organizing content (Moodle, 2024). Llamas et al. (2025) examined its institutional use at MSU–IIT, while Fojas (2022) and Almodiel (2022) examined gamification and discussions in the UPOU context. These sources show that Moodle can support several forms of academic work. For CdM LMS, the relevant comparison concerns the organization of coursework and the experience of completing institutional tasks.

2.3.2 Google Classroom

Mennuti (2024) described targeted assignments, rubrics, and analytics for grades, completion rates, and missing work in Google Classroom. The analytics described were associated with specified education editions or upgrades. These functions provide comparison points for assignment management and descriptive progress information. Their documentation does not establish the quality of the proposed CdM implementation or demonstrate that it will outperform Classroom.

2.3.3 Canvas and GitLab Integration

Schauer et al. (2024) implemented a framework connecting Canvas and GitLab to reduce fragmented workflows and support feedback. Its eight applications demonstrate that existing LMS and version-control platforms can be connected for educational use. This is a direct comparison for the integration concern in CdM's Git submission workflow. The technologies and automation described in that framework remain distinct from CdM's specified implementation.

2.3.4 GitSEED

Orvalho et al. (2024) presented GitSEED as a GitLab-backed automated assessment tool supporting programming and software engineering education. It uses repositories as part of submission and feedback workflows. Its approach provides a comparison for connecting coursework with version control, but its automated assessment capabilities extend beyond CdM's separate repository-submission and instructor-review process.

2.4 Synthesis of the Literature

The literature connects LMS adoption with useful academic functions, understandable interaction, and conditions supporting continued use. Alfalah (2023) and Dela Rosa (2023) address acceptance, while Quimod et al. (2023), Ramos et al. (2024), and Llamas et al. (2025) provide development and user-experience perspectives. Peralta (2024), Guico (2026) further connect coursework with communication and institutional support. These studies support examining connected workflows and appropriate access across CdM's complementary applications.

Programming and repository research address related but distinct academic processes. Messer et al. (2024), Asuncion et al. (2022), and Novak and Kermek (2024) inform checks against specified programming criteria. Schauer et al. (2024), Orvalho et al. (2024), and Patani et al. (2024) inform repository use and integration. The comparison supports keeping CodeLab's test-case assessment separate from Git-based coursework review, with instructors retaining responsibility for judgments outside the automated checks.

Gamification, personal study support, and collaboration concern different aspects of participation. Ahmed et al. (2025), Jack (2025), and Fojas (2022) provide evidence about rewards and engagement. Du et al. (2025), Heikkinen et al. (2023), Funa et al. (2023), and Pan et al. (2022) address regulation and active review. Oyarzun and Martin (2023), Almodiel (2022), and Mudau and Van den Berg (2023) emphasize interaction and facilitation. Their combined implication is to provide learning support with clear purposes and evaluate how the intended users experience it.

For the present study, these findings inform the development and descriptive evaluation of an LMS configured for ICS. Evidence from other institutions provides design guidance and comparison points. The study will establish its own findings about software quality and workflow suitability, while recognizing that user ratings and functional verification do not establish long-term educational effects.

2.5 Research Gap

The reviewed studies address digital learning and institutional LMS utilization but do not evaluate the proposed CdM LMS configuration for ICS at Colegio de Montalban. The research gap is contextual and evaluative: evidence is needed about how its combined workflows, roles, and complementary applications operate in the intended setting. One part of the gap concerns functional integration. CdM LMS will combine routine academic management with separate CodeLab and Git-submission workflows, learning-support tools, and moderated collaboration. Its evaluation must determine whether the selected functions produce the expected results and interact consistently within a shared information environment. This is a system-specific question; it is not a claim that no existing platform can provide similar integration.

A second part concerns academic coordination and information boundaries. The proposed expansion introduces an academic Super User, restricted course supervision, faculty organization, effective-dated timetable revisions, class exceptions, and audience-targeted events. The reviewed literature provides organizational reasons to consider coordination, but it does not establish whether CdM's particular implementation preserves schedule behavior, correct audiences, and restricted information access. These properties require requirements-based testing and evaluation of the implemented release.

A third part concerns role-appropriate access across applications. The web environment, complementary mobile client, and native Administrator Console have distinct responsibilities. The study will examine consistency in their applicable shared records and the quality of the tasks each interface supports, without assuming universal feature parity. Findings will identify the suitability and limitations of this configuration for ICS. The study will address these gaps through development, documented verification, and descriptive software-quality evaluation, producing evidence about the suitability and limitations of the ICS implementation.

CHAPTER 3 METHODOLOGY

3.1 Research Design

The study will combine iterative system development, requirements-based technical verification, and quantitative descriptive evaluation. Descriptive research systematically characterizes observed conditions and experiences without establishing causal effects (McCombes, 2023). Questionnaires will describe users' experiences with assigned tasks; controlled technical checks will examine correctness, access restrictions, record handling, performance, and integration. The two evidence types will be reported separately and considered together.

The Agile operational framework in Section 1.5 will guide development. Chapter 3 specifies how local requirements, evaluation participants, instruments, tests, and analysis will be organized. The evaluation will address six selected ISO/IEC 25010:2023 characteristics rather than claim assessment of the complete standard or ISO certification.

3.2 System and Project Overview

CdM LMS connects the principal Web Portal, complementary Expo/React Native application, and native Electron Administrator Console to Next.js backend workflows and PostgreSQL records. Students access approved coursework and their own academic and study information. Instructors manage assigned teaching activities. Super Users perform restricted academic coordination through the web; Administrators operate the native console. Program Head and Dean designations remain separate from those account roles.

The evaluation configuration will identify enabled modules, client builds, roles, and datasets. Existing functions and proposed coordination extensions will have separate implementation and test-status entries. Only demonstrated functions will be included in respondent tasks and reported as evaluated.

3.3 Local Requirements Gathering

Before finalizing the development backlog, the researchers will seek institutional permission to gather local requirements from relevant ICS students, instructors, and designated academic or

administrative personnel. Planned methods include guided interviews or small-group consultations and examination of available workflow descriptions. Topics will cover resource distribution, enrollment, submissions, grading, programming coursework, personal organization, faculty coordination, and device/network access.

With consent, the researchers will record de-identified notes, summarize recurring needs, and validate the resulting requirement statements with appropriate stakeholders. Each accepted requirement will identify its user, applicable platform, expected behavior, access restrictions, and acceptance criteria. These pre-development consultations are distinct from the post-development software-quality questionnaires.

3.4 Respondents

Respondents will comprise selected ICS students, ICS instructors, and qualified IT experts. Students must be officially enrolled in BSIT or BSCpE, have prior LMS experience, understand common academic tasks, and voluntarily perform the evaluation activities. Instructors must teach within ICS, have relevant LMS experience, and voluntarily complete their assigned teaching or coordination tasks. IT experts must have relevant qualifications, training, or experience in software development, testing, database management, cybersecurity, or system administration.

Students will evaluate enrolled coursework, submissions, personal grades, study tools, and collaboration that they have used. Instructors will evaluate assigned course management, assessment, grading, enrollment, and communication. Program Heads may evaluate designation-specific event publication. A Program Head designation will not be used as evidence of Super User authorization. Qualified experts may use controlled Super User and Administrator accounts to inspect oversight and console functions without unrestricted access to real student records.

The adviser and three proposed faculty validators will review instrument content separately from the evaluation sample. They will become respondents only if they independently meet the selection criteria. Target numbers for each respondent group will be justified after confirming eligible

participants and feasible task sessions; the final report will identify invited, participating, and valid responding counts separately.

3.5 Sampling Technique

Criterion-based purposive sampling will select participants whose experience and role allow them to evaluate the relevant functions. Memon et al. (2025) emphasize explicit and justified selection criteria in quantitative purposive sampling. Institutional enrollment alone will not qualify a participant to rate every platform or administrative function.

Following institutional permission, eligible participants will be invited through an approved recruitment process. The invitation will describe the study, tasks, participation conditions, and voluntary nature of involvement. The researchers will screen role and experience criteria, arrange task sessions, and record participation and nonresponse by group. Sample justification will reflect coverage of roles, functions, and feasible evaluation conditions; findings will not be represented as probability estimates for the entire ICS population.

3.6 Instruments and Data Collection

Two questionnaire sets will be prepared: a user instrument with Student and Instructor sections, and an IT-expert instrument for relevant technical, supervision, and administrative functions. Both will align with the selected quality characteristics. Items will describe one observable behavior and identify the task or function needed to answer them. Respondents will rate only the functions they have directly used.

Responses will use Strongly Agree (4), Agree (3), Disagree (2), and Strongly Disagree (1). A separate Not applicable or not observed option will receive no numeric score. The adviser and proposed validators will review clarity, relevance, role suitability, and alignment. A pilot with eligible participants outside the main sample will examine instructions and item interpretation. Content review and internal-consistency analysis of actual pilot responses will be distinguished; reviewer approval alone will not establish reliability.

Participants will receive orientation and consent information, use assigned test accounts where applicable, and perform a documented task sequence before completing the questionnaire. The researchers will record role, platform/build, task completion, and encountered issues. Technical checks will use a separate record of setup, expected and actual results, errors, measurements, and retest outcomes.

Table 2. Role appropriate evaluation tasks

Evaluator	Account/platform	Representative tasks
ICS students	Student; web/selected mobile	Approved coursework, own grades, notes/tasks, study tools and Knowledge Exchange
ICS instructors	Instructor; web/selected mobile	Assigned materials, submissions, grading, enrollment and schedules
Eligible Program Heads	Instructor designation; web	Permitted faculty/program event publication
Qualified IT experts	Controlled Super User and Administrator accounts	Restricted supervision, protected console operations and direct technical checks

3.7 Technical Testing and Acceptance

A test matrix will map requirements to selected quality characteristics, account roles, platforms, datasets, and expected outcomes. Each case will record the application revision, enabled feature, setup, action, result, error evidence, and retest status. Calculation checks will compare outputs with manually verified expected records. Security cases will invoke prohibited reads/writes directly rather than rely on hidden interface buttons.

The researchers will define workload and dataset sizes before performance testing. Logged-in sessions, request rates, simultaneous programming executions, and upload sizes will be recorded separately; a registered-account count will not establish concurrent capacity. Repeated timing checks will distinguish ordinary retrieval, grade computation, calendar queries, file transfer, and CodeLab execution. Thresholds will be approved and recorded before judging performance.

Reliability checks will examine saved records after refresh, interrupted requests, retries, duplicate conversion, stale saves, and relevant concurrent updates. Compatibility checks will verify shared records through permitted clients and observe the Administrator Console alongside selected workstation software. Migration and restoration rehearsals will use an isolated database. Unexecuted

or failed cases will remain visible in the evaluation record; fresh evidence will support any reported pass.

Table 3. Technical evidence for selected quality characteristics

Characteristic	Checks	Evidence
Functional Suitability	Controlled computations, submissions, schedules and conversions	Expected/actual output and coverage
Interaction Capability	Role-appropriate tasks, navigation and feedback	Completion observations and ratings
Security	Prohibited API/action requests, private-record isolation	Access decisions and audit evidence
Performance Efficiency	Recorded workloads, timings and resource use	Measurements with setup and repetitions
Reliability	Interrupt/retry, duplicate/stale save, persistence	Record consistency and recovery outcomes
Compatibility	Permitted shared records; console co-existence	Client consistency and workstation observations

3.8 Data Analysis and Statistical Treatment

Questionnaire responses will be tabulated by respondent group, item, and quality characteristic. Frequencies and percentages will describe response distributions. The valid count for each item will exclude unanswered and Not applicable or not observed responses; those exclusions will also be reported. N/A will not be treated as zero, disagreement, or a passing result.

For comparability with the coded scale, an item mean will be calculated as the sum of valid response scores divided by the valid response count. Equivalently, the frequency-weighted mean is (4f4 + 3f3 + 2f2 + f1) / (f4 + f3 + f2 + f1). Frequency weighting represents counts of responses, not different importance weights for users or quality characteristics. Response distributions and valid counts will accompany means; median summaries may be reported to retain visibility of the ordered categories.

Characteristic summaries will first be computed for each respondent from the applicable items, then summarized within the Student, Instructor, or expert group. The number of applicable items and contributing respondents will be reported. Results from different roles or questionnaire versions will not be pooled into one score unless item equivalence and weighting are explicitly

justified. Numeric results will be reported on the 1–4 scale without unapproved labels such as highly acceptable.

Technical results will be reported by test case and characteristic. Correctness and access cases will identify expected/actual outcomes and pass, fail, or not executed status. Timings will identify workload, repetitions, median and observed slow cases; resource observations will identify the measurement conditions. Favorable questionnaire means will not override failed authorization, calculation, or persistence tests. The study will use descriptive summaries and will not claim causal educational effects or population-wide statistical representation.

3.9 Participant Protection and Information Handling

The researchers will obtain applicable institutional permission before recruitment or access to restricted records. Participants will receive information about purpose, tasks, duration, data use, and voluntary participation. Declining or withdrawing will not affect grades, employment, or access to ordinary institutional services. Consent will be obtained before task participation and questionnaire collection.

Evaluation identifiers will replace participant names in analysis files. The researchers will use controlled accounts and synthetic or appropriately de-identified academic records for restricted tests. Only authorized members of the research team and approved reviewers will access identifiable materials. Storage location, access controls, retention period, and deletion arrangements will be specified in the approved protocol. Public reports will present aggregate results and avoid exposing credentials, repository tokens, or personal academic records.

3.10 Hardware and Deployment Requirements

Equipment specifications below are planning estimates for development and a small evaluation pilot. They are not experimentally established minimums or verified concurrent capacity. The actual device and hosting configuration will be recorded before task sessions.

Table 4. Estimated development and client hardware

Equipment	Estimated baseline	Recommended	Purpose
Development workstation	4 cores; 16 GB RAM; 50 GB free SSD	6+ cores; 32 GB RAM; 100 GB free SSD	Build clients and run local checks
Web computer	Dual core; 4 GB RAM; 1366×768; 1 GB free	4 cores; 8 GB RAM; 1080p; 5 GB free	Academic and coordination tasks
Android device	Build-compatible; about 4 GB RAM; 1 GB free	6+ GB RAM; 2 GB free	Enabled mobile tasks
iOS device if evaluated	Compatible iPhone/iPad; 1 GB free	Supported device; 2 GB free	Verified iOS build only
Administrator workstation	64-bit dual core; 4 GB RAM; 2 GB free	4 cores; 8 GB RAM; 5 GB free	Native Console and exports
Network equipment	Reliable router/AP and connectivity	Wired laboratory links and sufficient Wi-Fi capacity	Connect authorized clients/services

Storage allowances cover dependencies, temporary files, cache, and downloads rather than measured installer sizes. Emulators, containers, and datasets may need more development space. Enabled 3D activities require a tested WebGL 2-capable browser/graphics configuration; dedicated graphics is not assumed for ordinary LMS tasks.

Table 5. Proposed pilot hosting and network configuration

Component	Initial estimate	Condition to verify
Application service	2 vCPU / 4 GB RAM on a conventional server	For serverless hosting, record plan/function limits instead
PostgreSQL	2 vCPU / 4 GB RAM; 10–20 GB initial records/index storage	Connection limits, growth and query behavior
Persistent file storage	20–50 GB initial object-storage allowance	Materials, submissions, snapshots and retention
Code execution if self-hosted	Separate 4 vCPU / 8 GB RAM; 30–50 GB storage	Languages, isolation, test limits and execution concurrency
Backups	Separate protected storage	Retention capacity and demonstrated restoration
Individual connection	About 5 Mbps down / 2 Mbps up; prefer 10/5	Measured latency, contention and file sizes
Shared lab connection	About 100 Mbps down / 20–50 Mbps up for an initial 20–30-device session	Actual simultaneous workload and connection stability

The initial sizing assumption is approximately 25–50 simultaneously active ordinary-task users, not a research sample size or proven capacity. Code executions will be limited and measured separately. Lower bandwidth may support basic retrieval, while uploads and concurrent activity may

need more. Managed-service quotas and the demonstrated workload take precedence over these estimates.

3.11 Software and Client Requirements

The software record identifies the current development stack and proposed evaluation targets. It will be frozen against the actual release revision; framework support alone does not establish CdM client compatibility.

Table 6. Software environment and release configuration

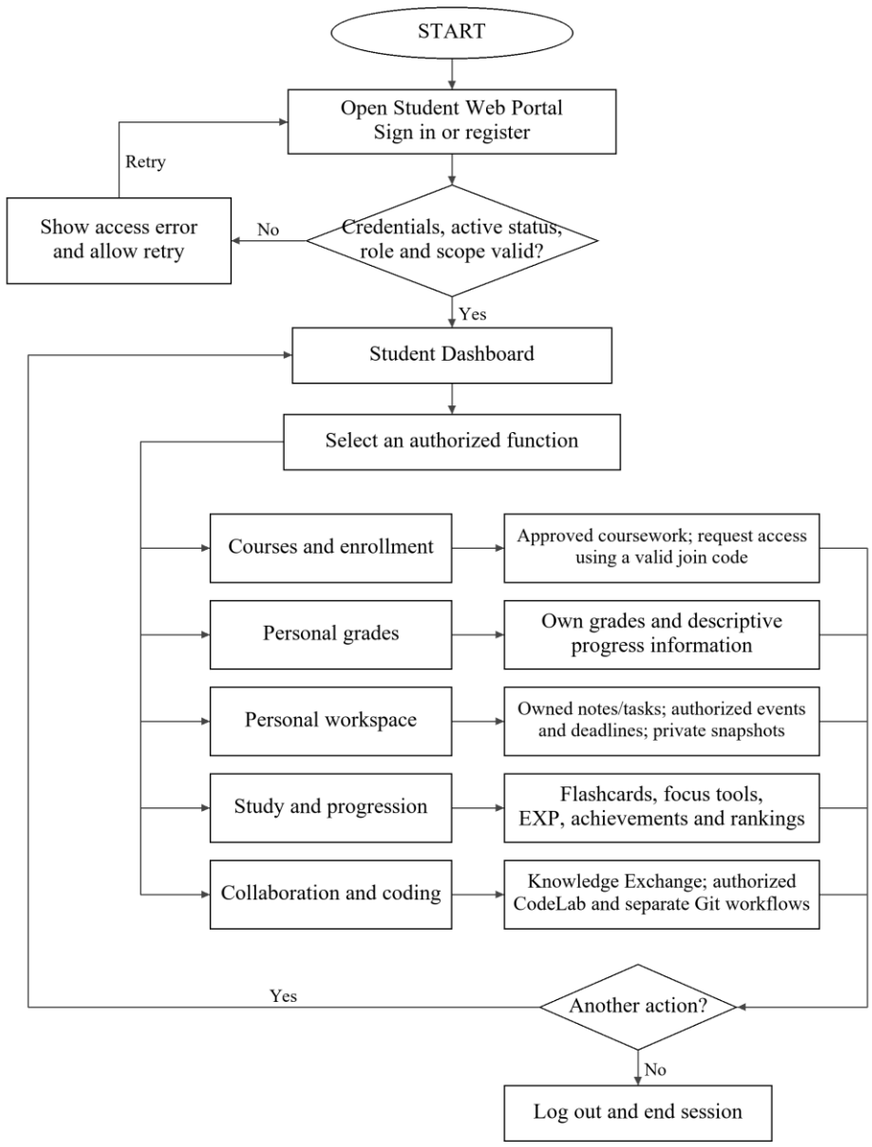
Software/service	Recorded choice	Purpose/qualification
Development OS	Supported Windows/macOS/Linux environment	Platform-specific build requirements apply
Node and workspace tools	Supported Node 22 LTS patch ≥22.12; pnpm 10.32.1; Turborepo	Record exact tested runtime/build versions
Web interface/backend	Next.js 16.2.0; React 19.2.4	Principal portal and backend workflows
Typing and styles	TypeScript 5.9.3; Tailwind CSS 4.2.2	Application implementation
Database	PostgreSQL; 16 proposed as isolated test baseline	Verify the deployed LMS version separately
Database access	Prisma 7.9.1 and PostgreSQL adapter/driver	Schema access and reviewed migrations
Mobile	Expo 55.0.31; React Native 0.83.10; Expo Router	Evaluate selected enabled functions
Desktop	Electron 41.0.3; Electron Builder	Native Administrator Console
Browsers	Current Chrome/Edge initially; Firefox/Safari where tested	Record exact versions and enabled activities
CodeLab execution	Configured Judge0-compatible endpoint	Verify service version, languages, isolation and limits
Repository coursework	Git and GitHub integration	Repository authorization is separate from LMS login
Files and email	Persistent object storage; configured account email service	Verify effective upload limits and supported delivery
Testing	Playwright 1.63.0; TypeScript/Node and harness checks	Fresh functional, access and browser evidence
Observation and analysis	Developer tools, logs, database observations; Excel or LibreOffice Calc proposed	Select workload tools and record actual versions

Proposed evaluation targets are Windows 11 x64, Android 10+ and iOS 16+, conditional on verified builds.

3.12 Role Based Flowcharts

3.12.1 Student View

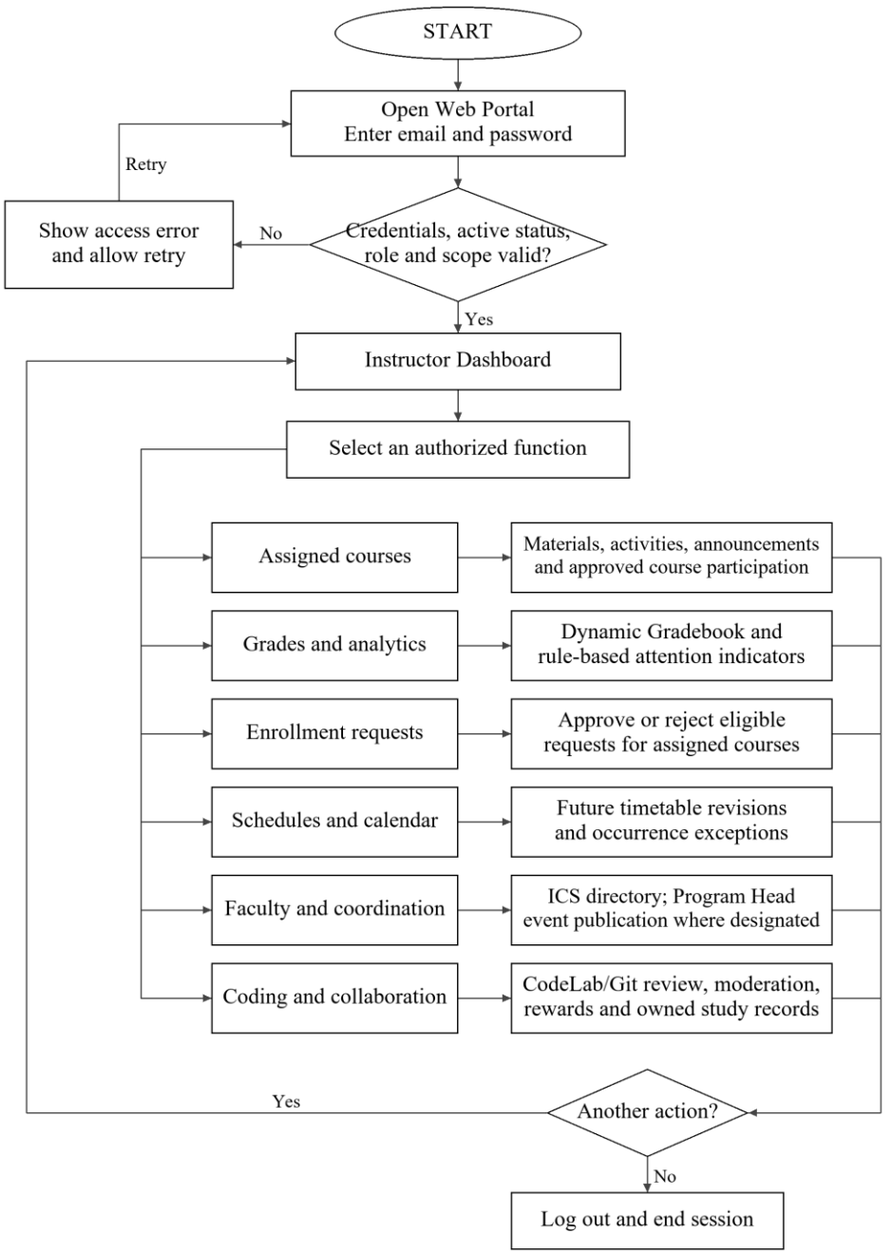
Figure 3. Student View navigation and access flow



Existing students choose the institute portal and enter email/password; new students use the supported registration process. Backend checks establish account, role and institute access before opening the dashboard. Students access approved coursework and their own grades, personal records, study tools, and authorized collaboration/programming tasks. Calendar access comprises readable saved events and coursework deadlines; faculty-only events and recurring student timetable projection are excluded. Each protected operation is validated by its source workflow, and the session can return to the dashboard or end through logout.

3.12.2 Instructor View

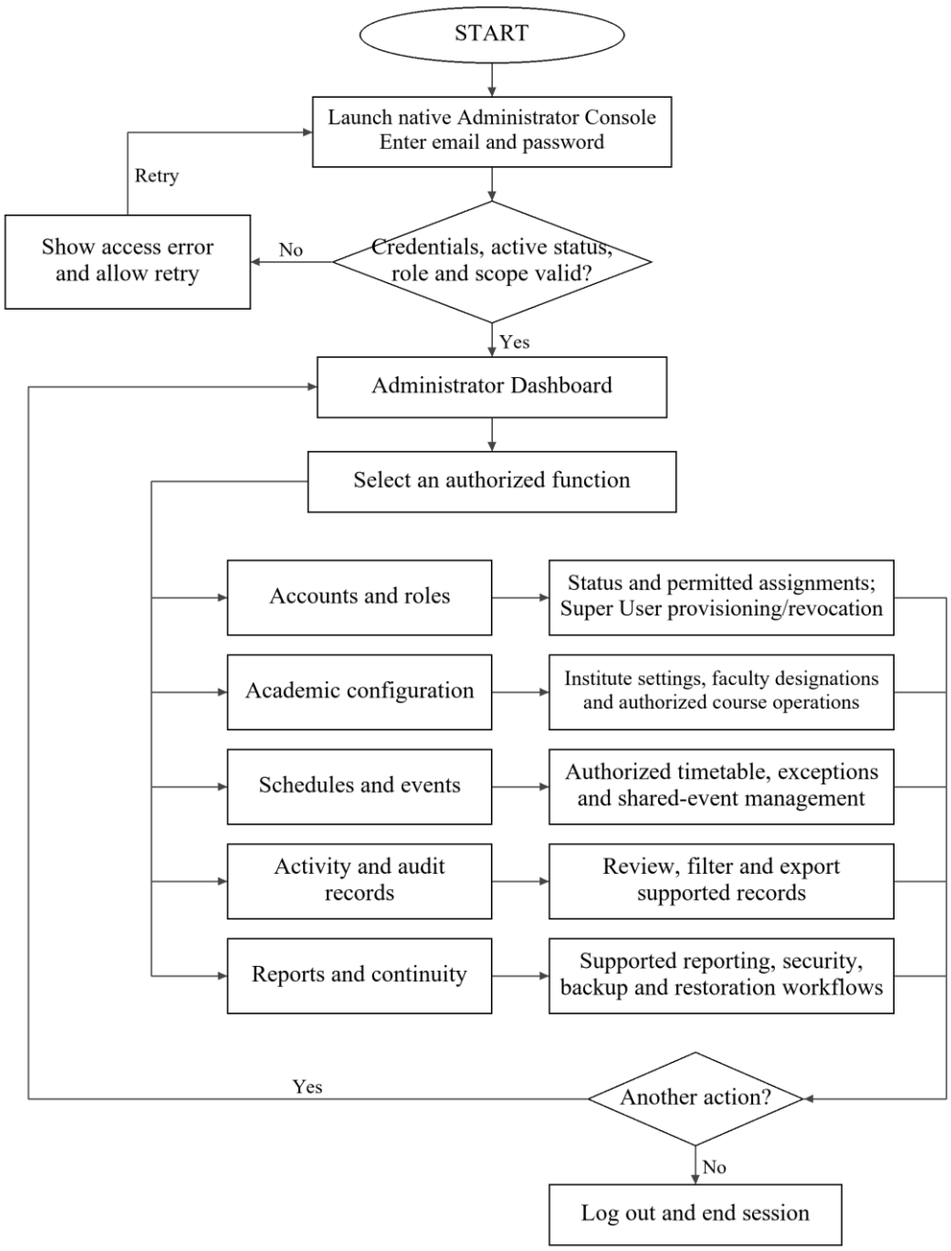
Figure 4. Instructor View navigation and access flow



Instructors authenticate through the portal and operate on assigned courses. They manage materials, assessments, submissions, grades and enrollment decisions; inspect descriptive analytics and rule-based attention indicators; and use authorized schedules, calendar and directory functions. Program Head publication is available only with a valid designation. Scheduling changes preserve earlier revisions, while occurrence exceptions affect permitted future classes. Programming results and repository review remain distinct. Moderation, rewards and owned study functions follow their defined permissions.

3.12.3 Administrator View

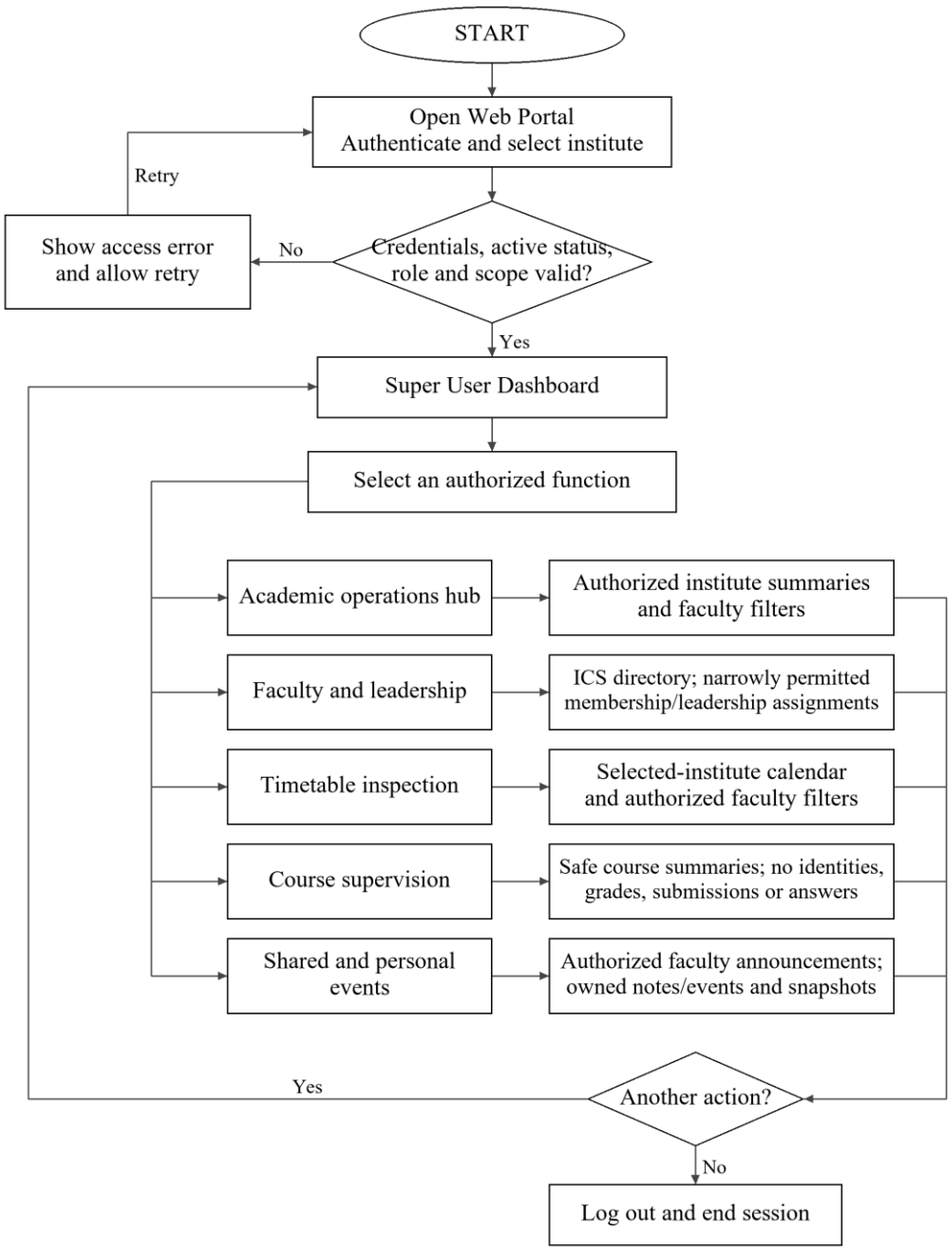
Figure 5. Administrator View navigation and access flow



The Administrator launches the native Console and enters email/password. The backend validates active status and Administrator authority before providing protected operations. Supported functions include accounts, roles, instructor assignments, Super User provisioning, institute/faculty configuration, course and calendar administration, log review, and demonstrated reporting/continuity tools. Audit records are reviewed rather than freely rewritten. Confirmation and reauthentication apply where required by the implemented operation. Additional browser/mobile administration is not assumed by this evaluation.

3.12.4 Super User View

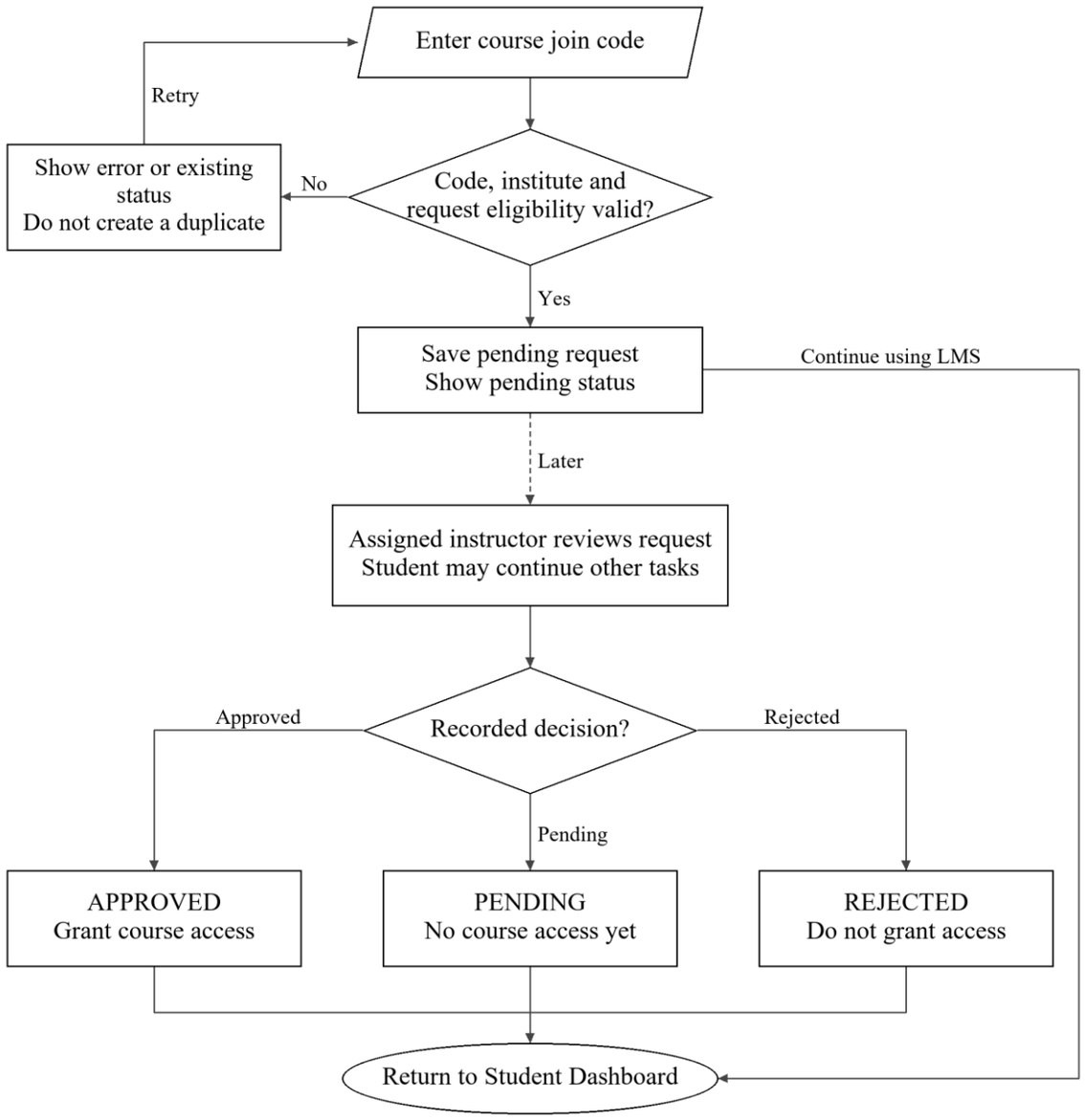
Figure 6. Super User View navigation and access flow



The Super User authenticates through the Web Portal and selects a supported institute, validated independently of the account's home institute. The academic hub provides permitted faculty information, timetable filters, narrow leadership assignments, shared announcements, and safe course summaries. Supervision excludes student identities, grades, submissions, attempts and answers. The role does not permit grading, assessment participation, course schedule editing, or generic Administrator tools. Owned notes/events remain private. Users return to the hub for another action or log out.

3.12.5 Enrollment Request Flow

Figure 7. Course enrollment request and instructor decision

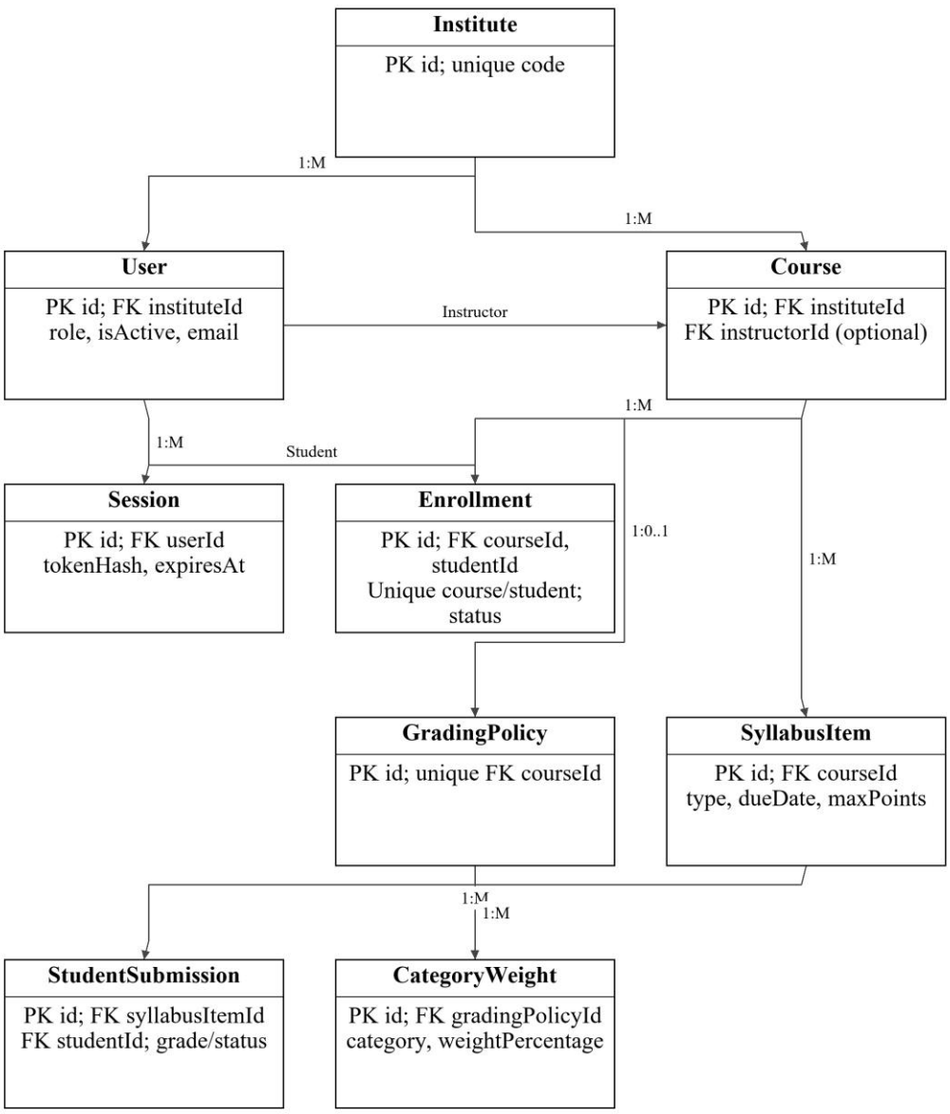


A valid join code does not immediately grant course access. The backend checks the institute and existing status, rejects invalid/duplicate requests, and records eligible requests as pending. The student may continue using other modules while waiting. The assigned instructor records approval or rejection; only approval grants enrolled-course access. The dashed review transition represents a later instructor action rather than a synchronous student operation. Expired sessions return to authentication, and failed source operations provide feedback without implying that unsaved changes succeeded.

3.13 Entity Relationship Diagrams

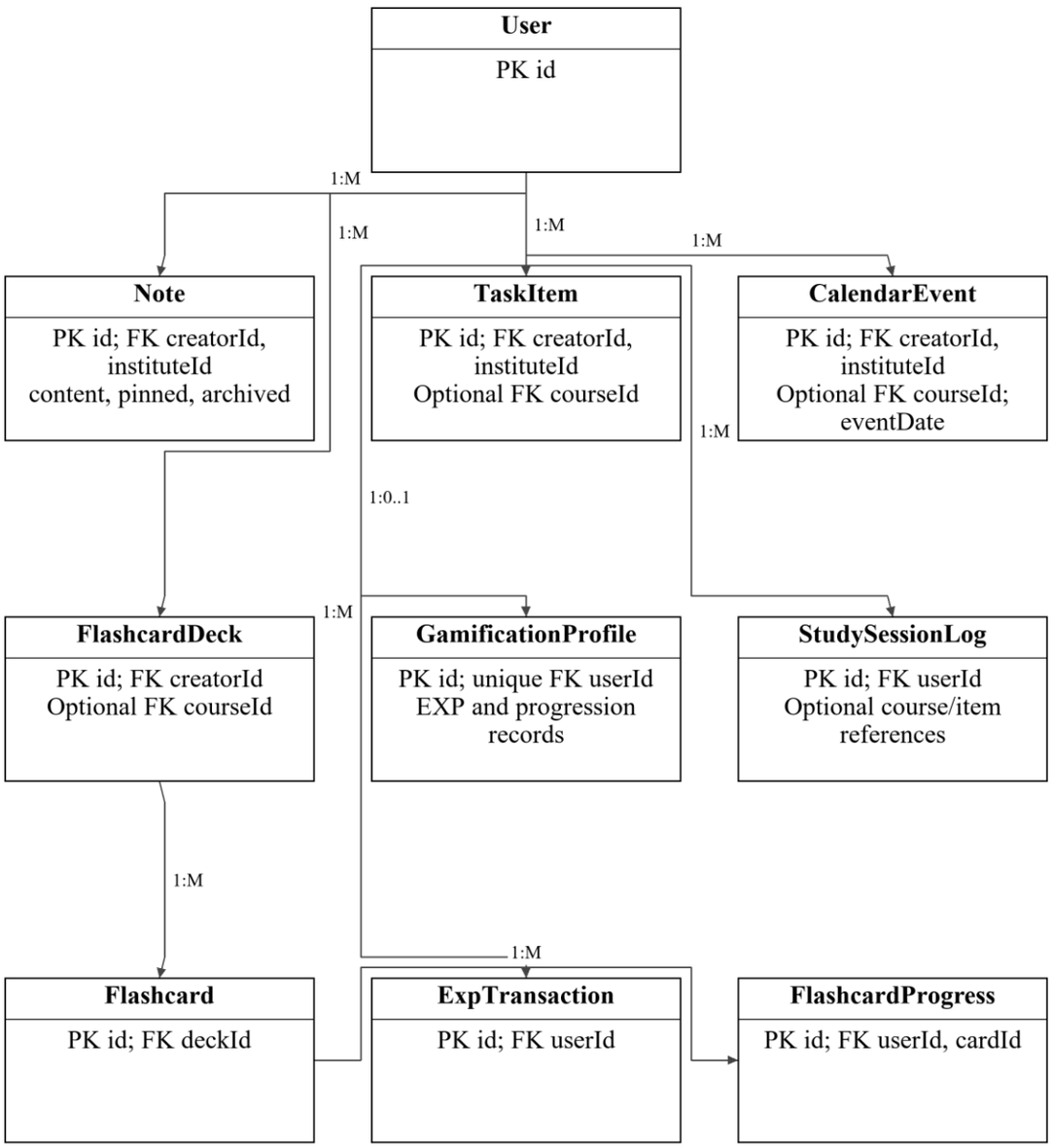
Figures 8–11 summarize selected current Prisma relationships and the agreed coordination target. User stores account roles, Enrollment links students and courses, and grading uses applicable submission records and policies. PK/FK denote primary/foreign keys. These scoped figures must agree with the frozen release schema.

Figure 8. Selected existing account and academic relationships



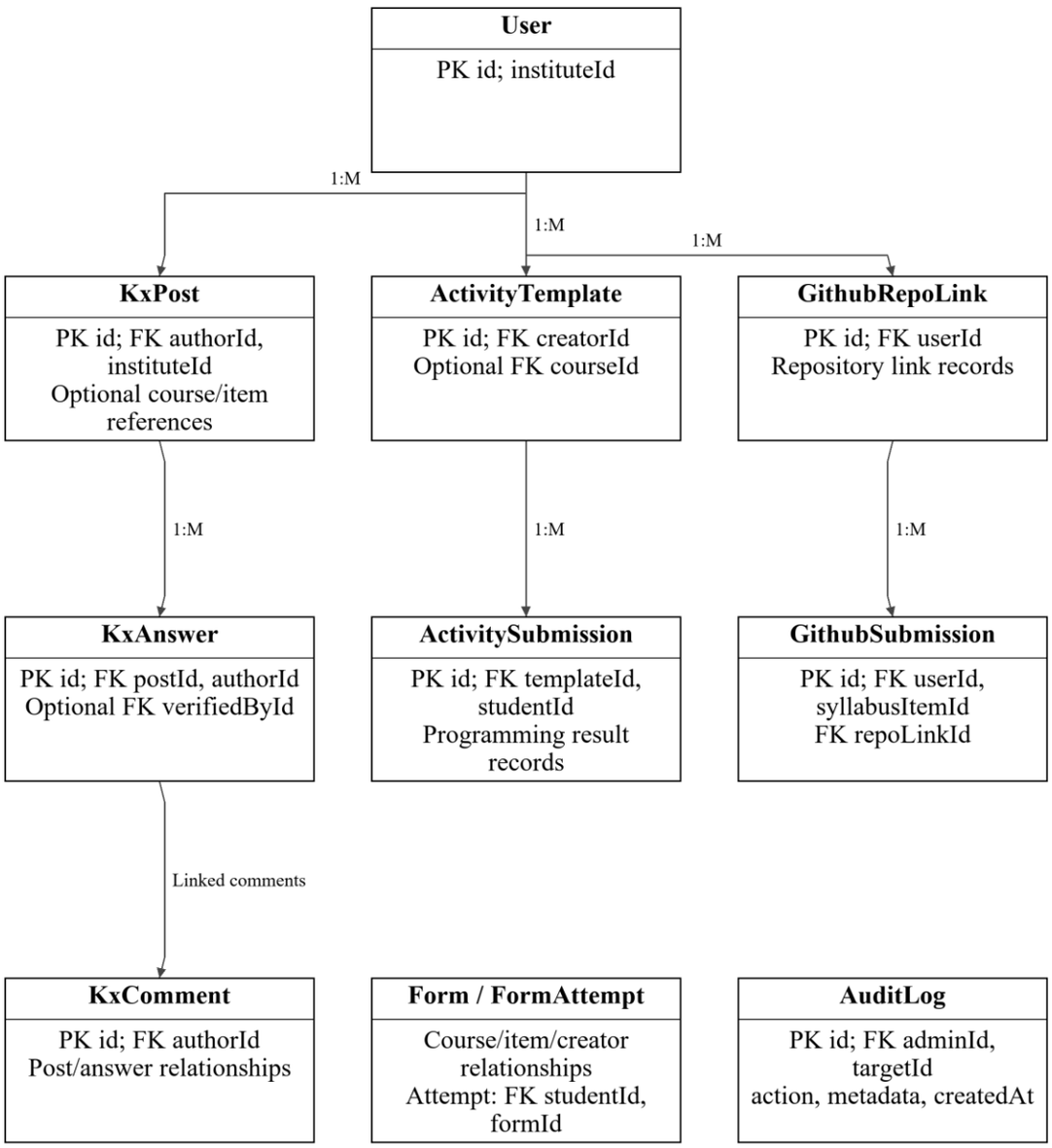
Relationships identify the principal references in this functional group. Optional fields permit the stated unassigned or unlinked state; access policy is enforced separately from database relationships. Additional supporting records, indexes, constraints and assessments must be checked against the frozen evaluation schema.

Figure 9. Selected existing study and progression relationships



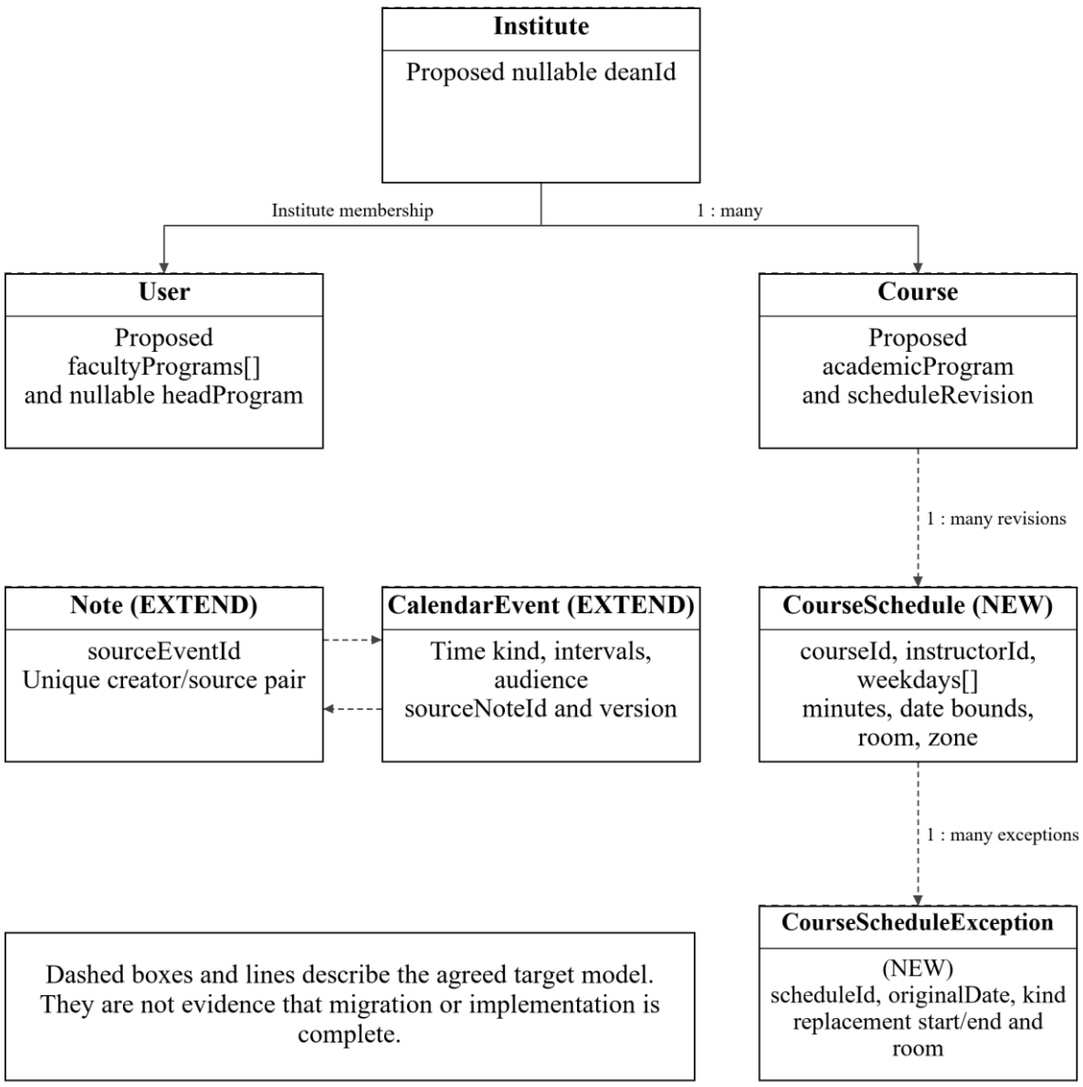
Relationships identify the principal references in this functional group. Optional fields permit the stated unassigned or unlinked state; access policy is enforced separately from database relationships. Additional supporting records, indexes, constraints and assessments must be checked against the frozen evaluation schema.

Figure 10. Selected existing programming and forum relationships



Relationships identify the principal references in this functional group. Optional fields permit the stated unassigned or unlinked state; access policy is enforced separately from database relationships. Additional supporting records, indexes, constraints and assessments must be checked against the frozen evaluation schema.

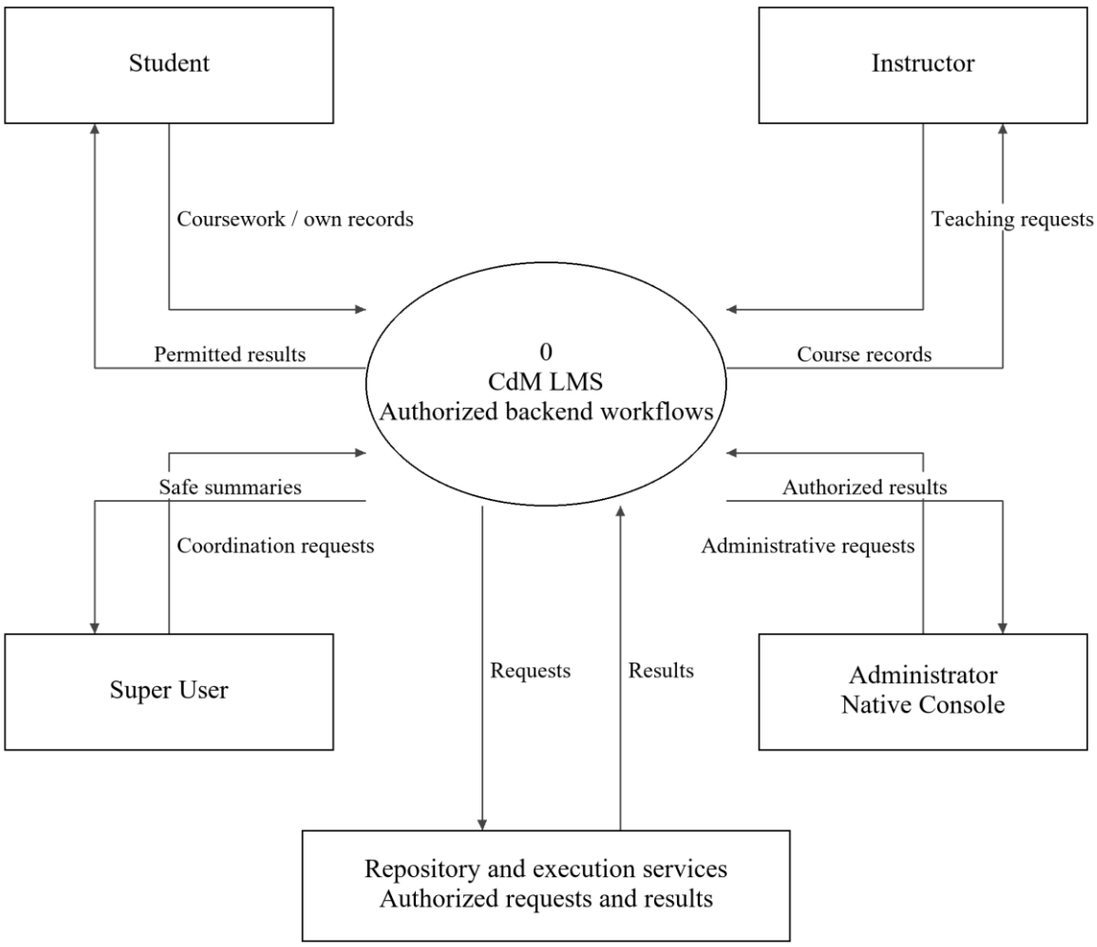
Figure 11. Proposed academic coordination relationships



The target model introduces CourseSchedule and CourseScheduleException and extends Institute, User, Course, CalendarEvent and Note. Schedule revisions store weekdays, minutes, date bounds, timezone, room and attribution; exceptions identify original occurrences. Event audiences/time fields and source links support private snapshots. One head per program, head membership, unique conversion links, transactions and version comparisons enforce the agreed integrity rules. Dashed items are proposed until reviewed migrations and feature verification establish them.

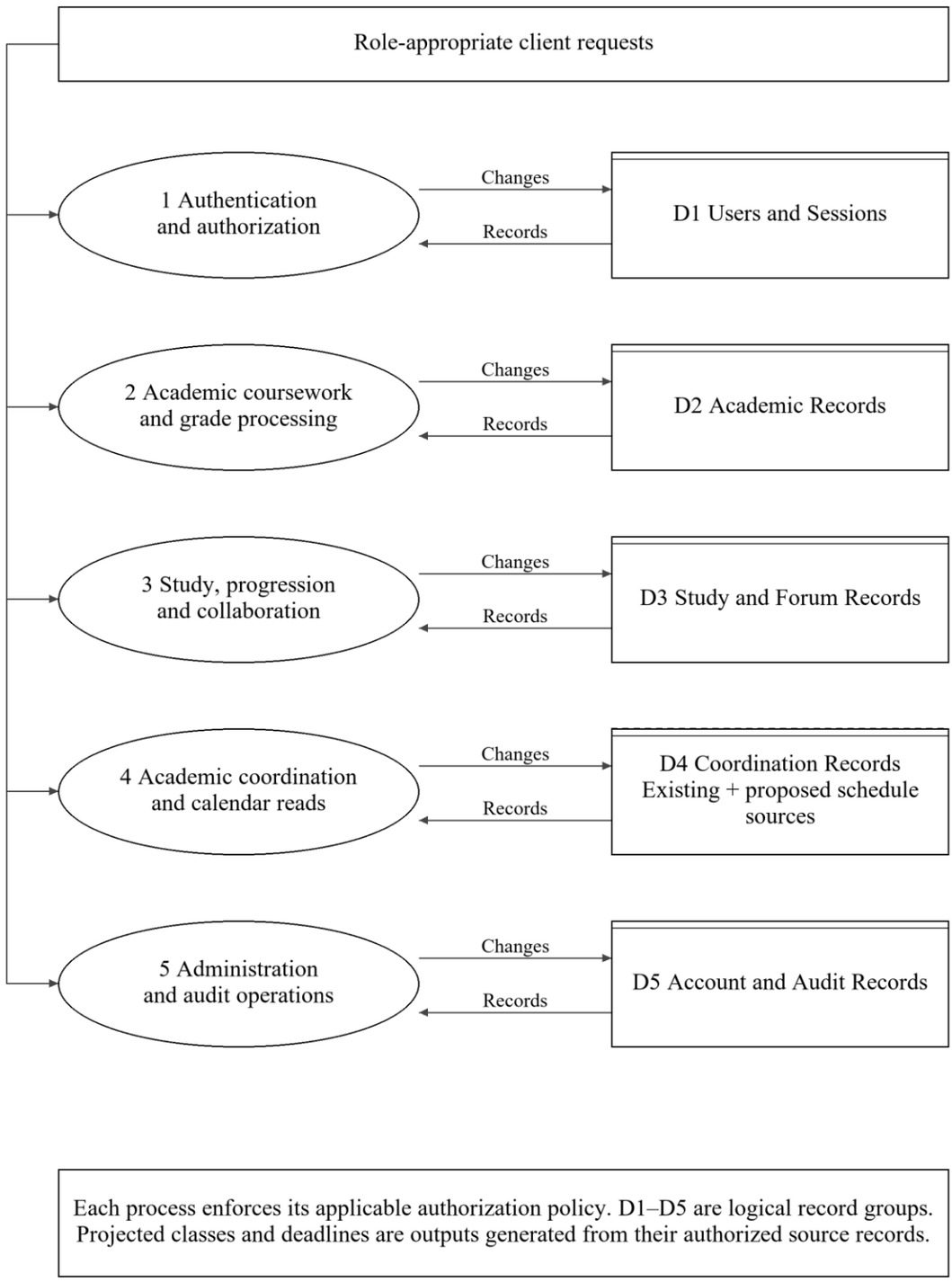
3.14 Data Flow Diagrams

Figure 12. Context diagram of authorized CdM LMS information exchange



The context diagram identifies the four role categories as external actors exchanging requests and permitted results with CdM LMS. Repository and execution services support their respective programming workflows. Data exchange occurs through authorized backend operations rather than direct client database access. Inputs, returned records and external-service results are bounded by role, institute, ownership and course relationships.

Figure 13. Level 1 functional data flow and principal record groups



Level 1 decomposes the system into authentication/authorization, academic coursework and grading, personal study/progression/collaboration, academic coordination, and administration. D1–D5 represent logical record groups rather than additional database tables. D4 includes existing event records and the proposed coordination extension. Generated classes and syllabus deadlines are calendar read outputs, not routine persisted event rows. Each functional process validates its requests and retrieves or changes only authorized records.

3.15 Release Configuration and Status Record

Before recruitment, the researchers will freeze the release revision and record client builds, schema/migration state, endpoint configuration, enabled languages, repository provider, accepted files, device/browser matrix, network conditions, role accounts, datasets, and known limitations. Implementation and test status will be recorded separately. Secrets will remain outside manuscript examples and public repositories.

Table 7. Feature status categories for the evaluation record

Category	Required release evidence
Existing module	Source/workflow identified; actual coverage and role access verified
Planned extension	Requirement and target model recorded; not reported as complete
Tested function	Fresh expected/actual results, setup and retest status
Evaluation enabled	Reviewed function/build included in assigned participant tasks

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